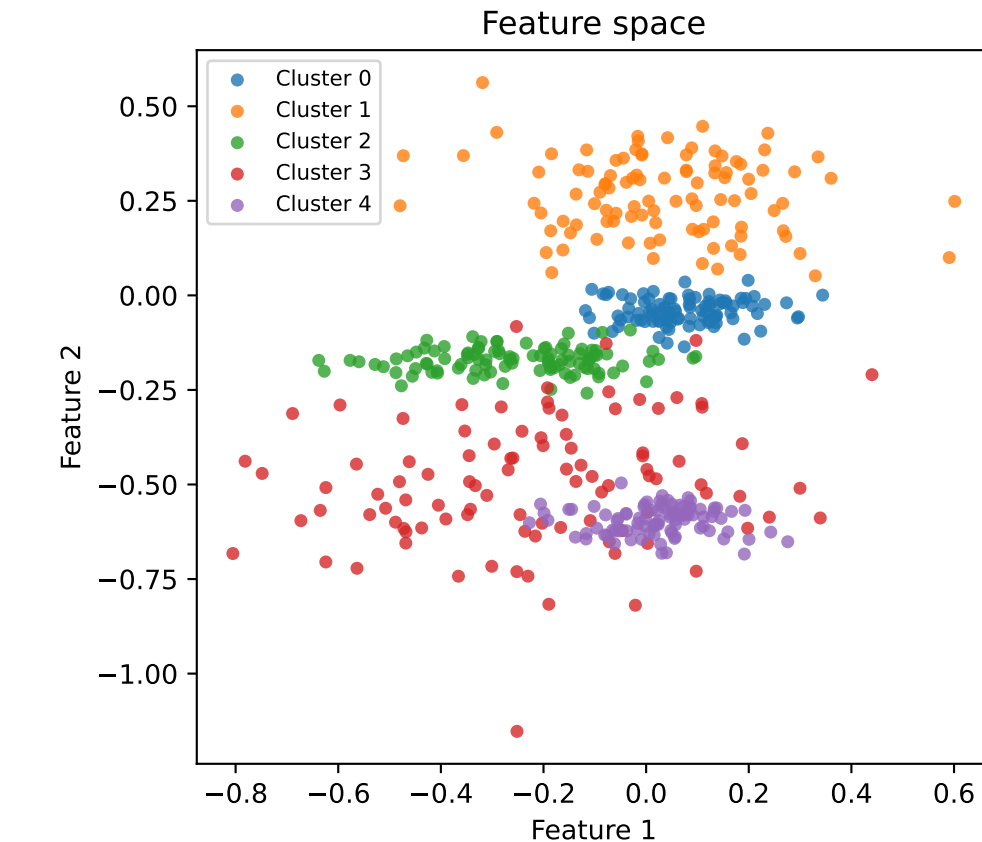
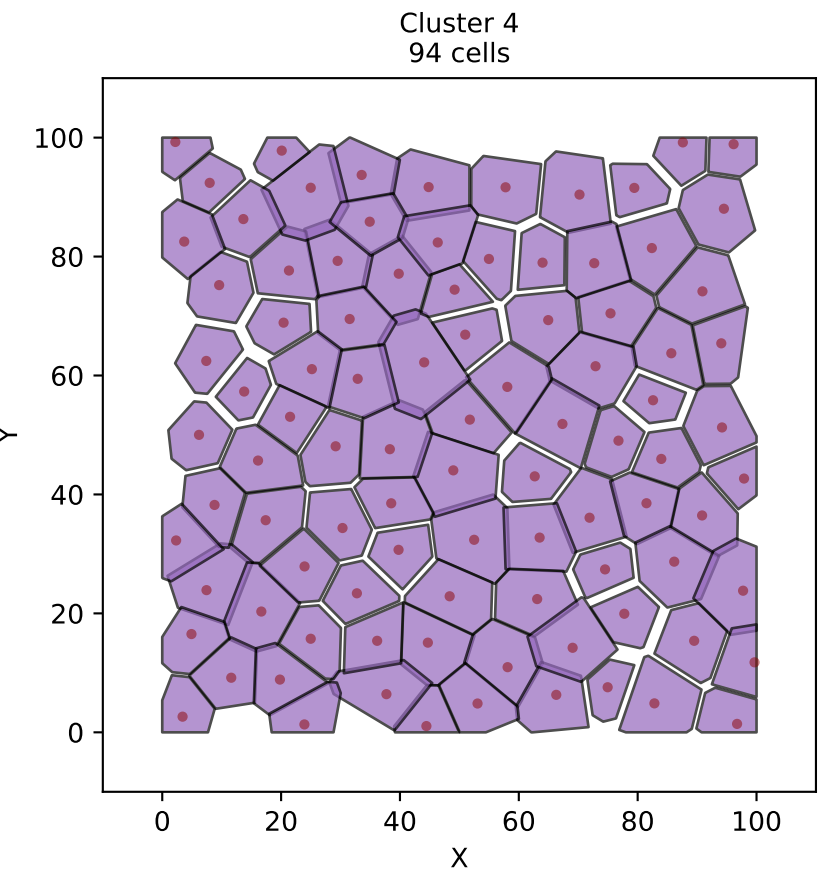
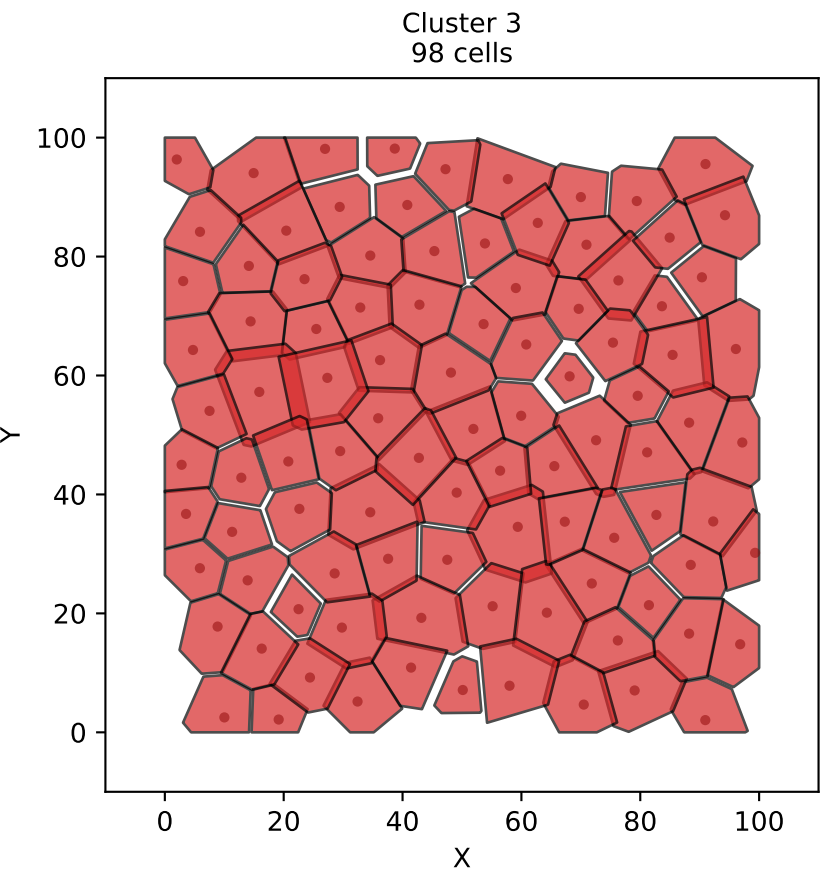
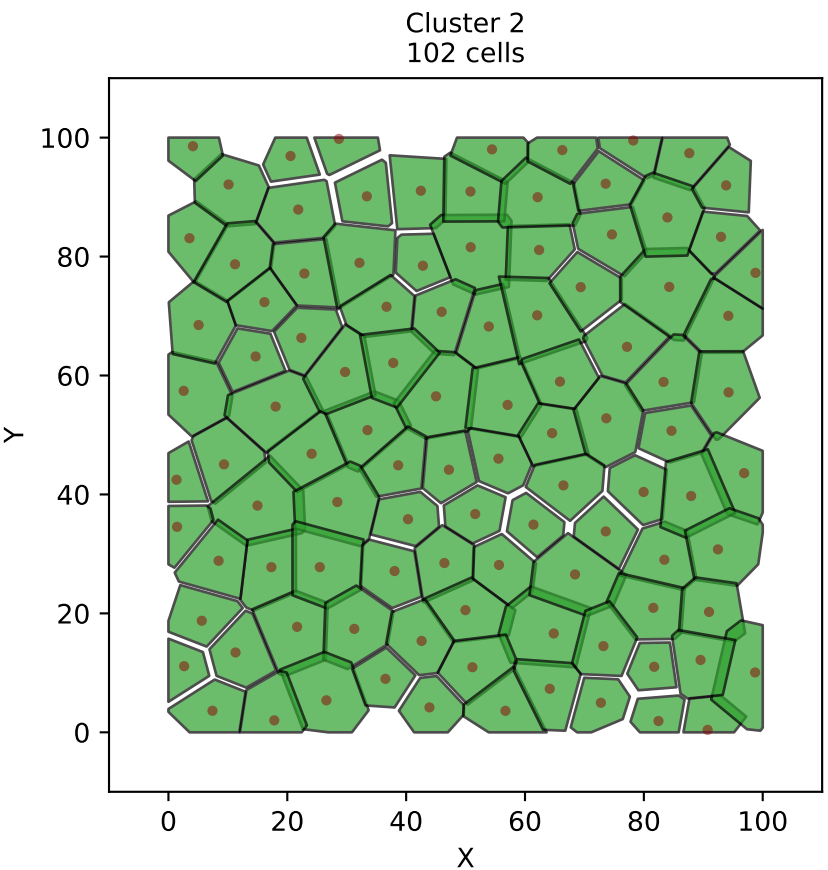
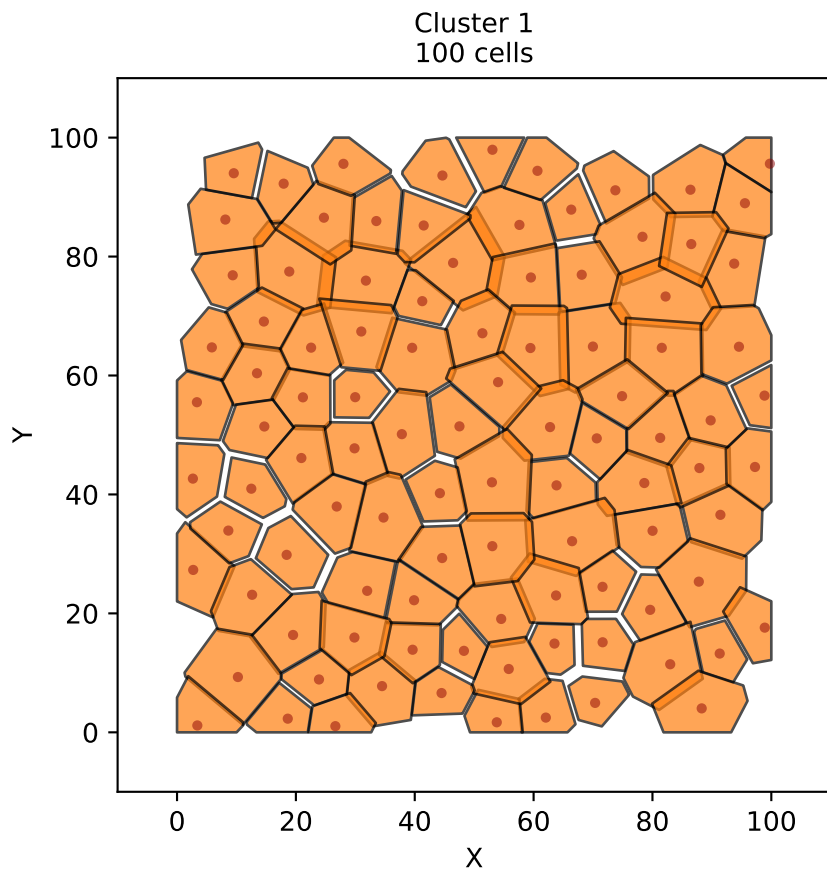
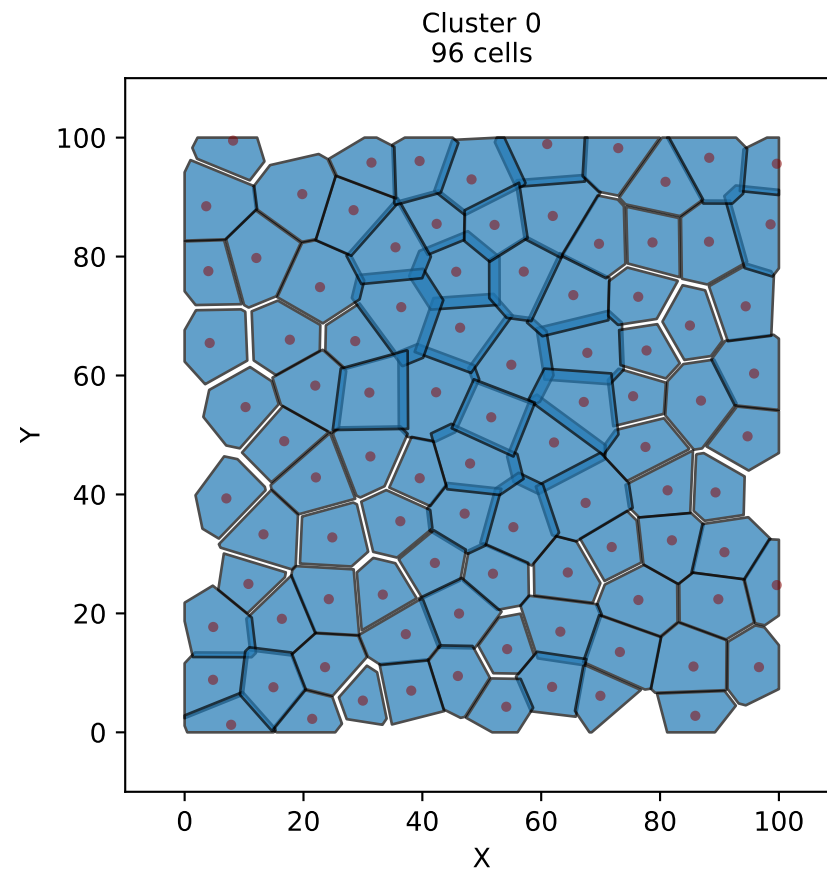
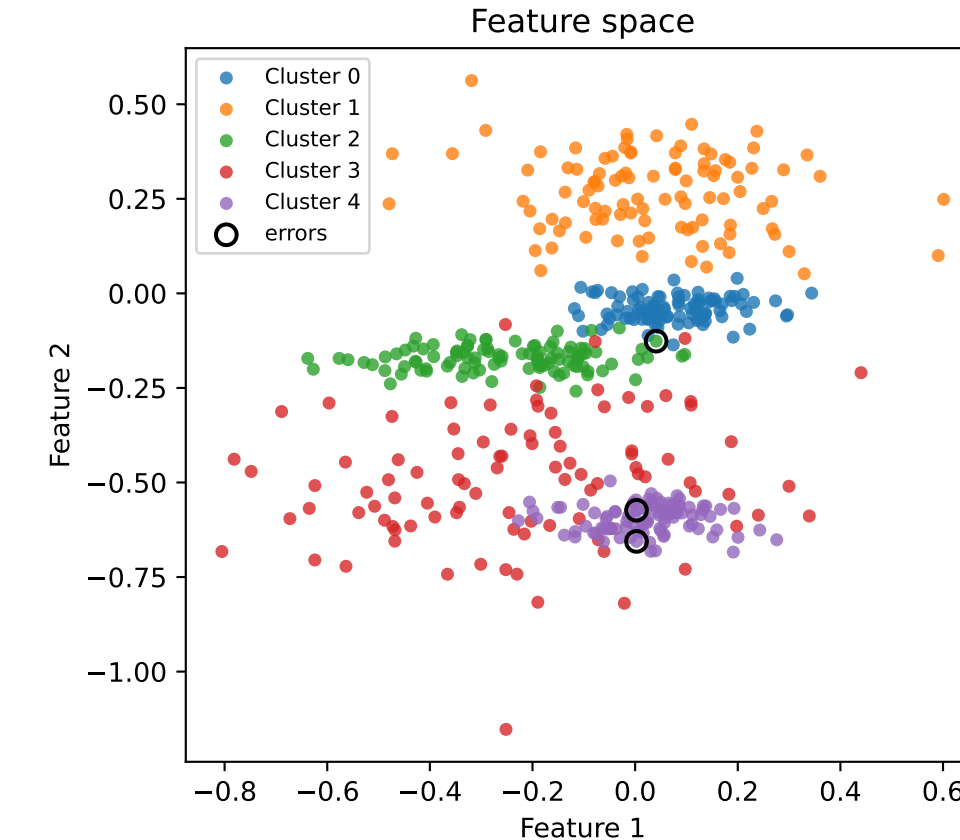
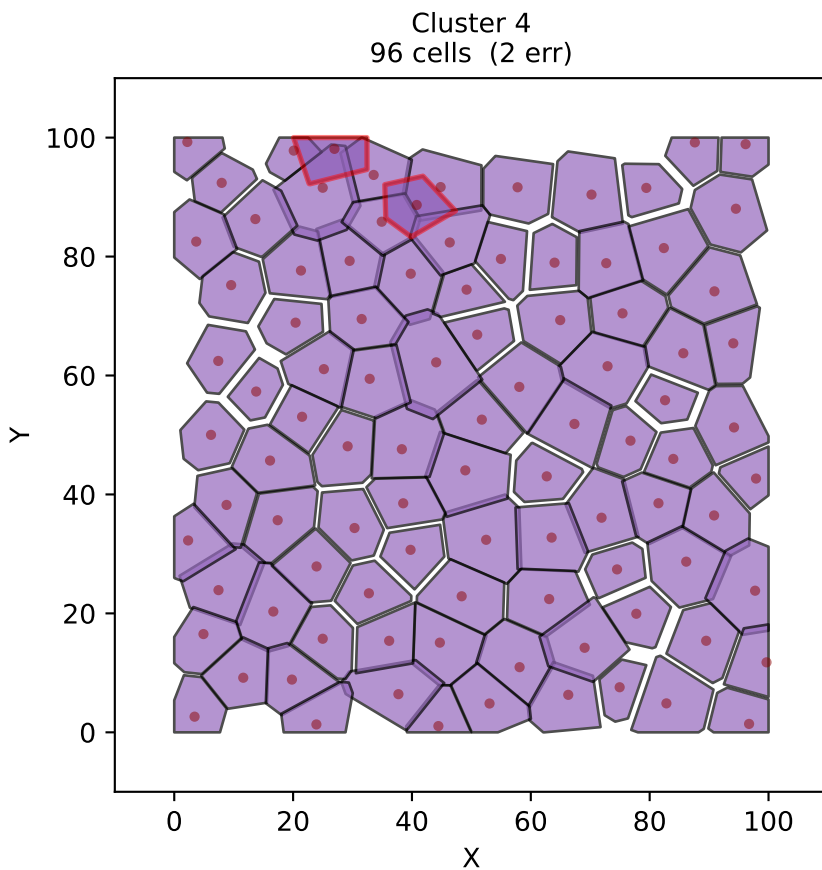
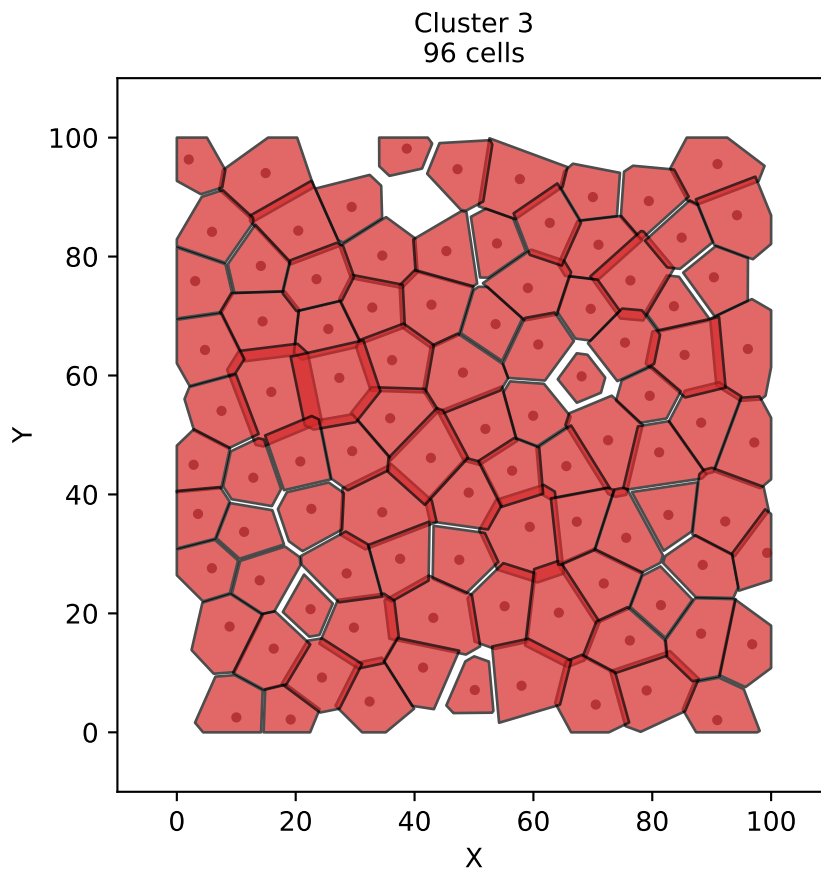
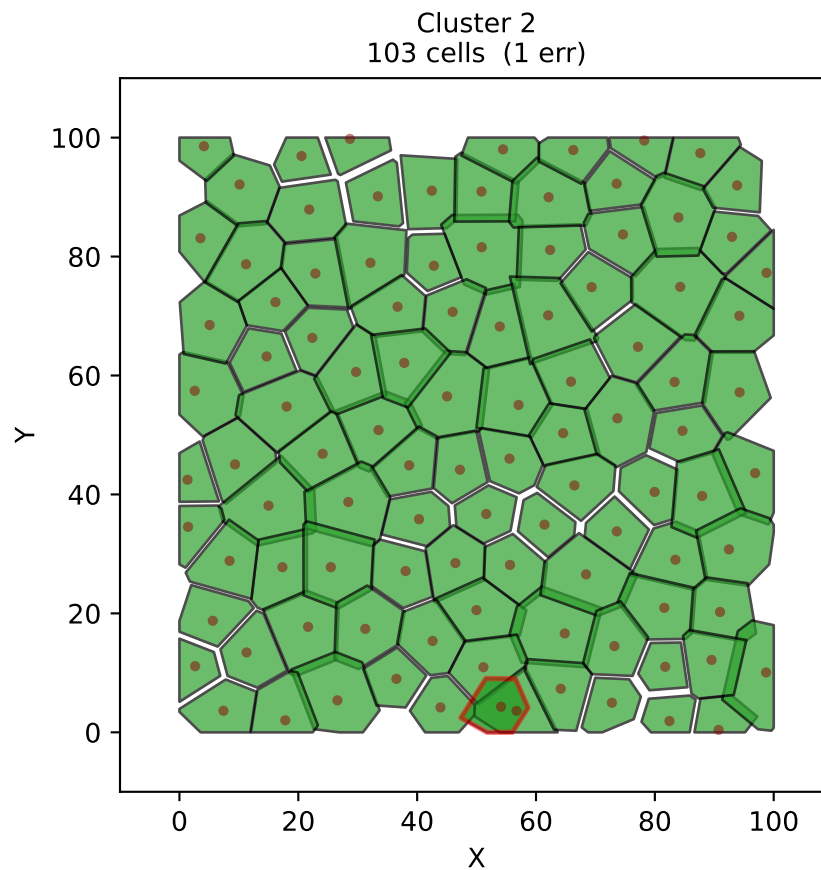
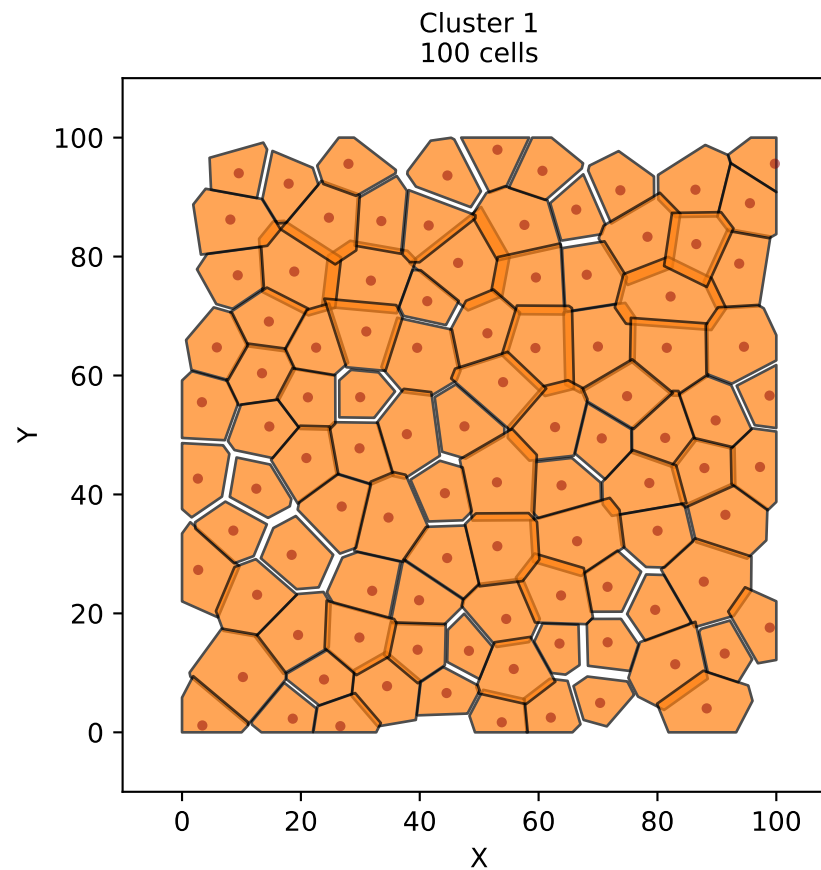
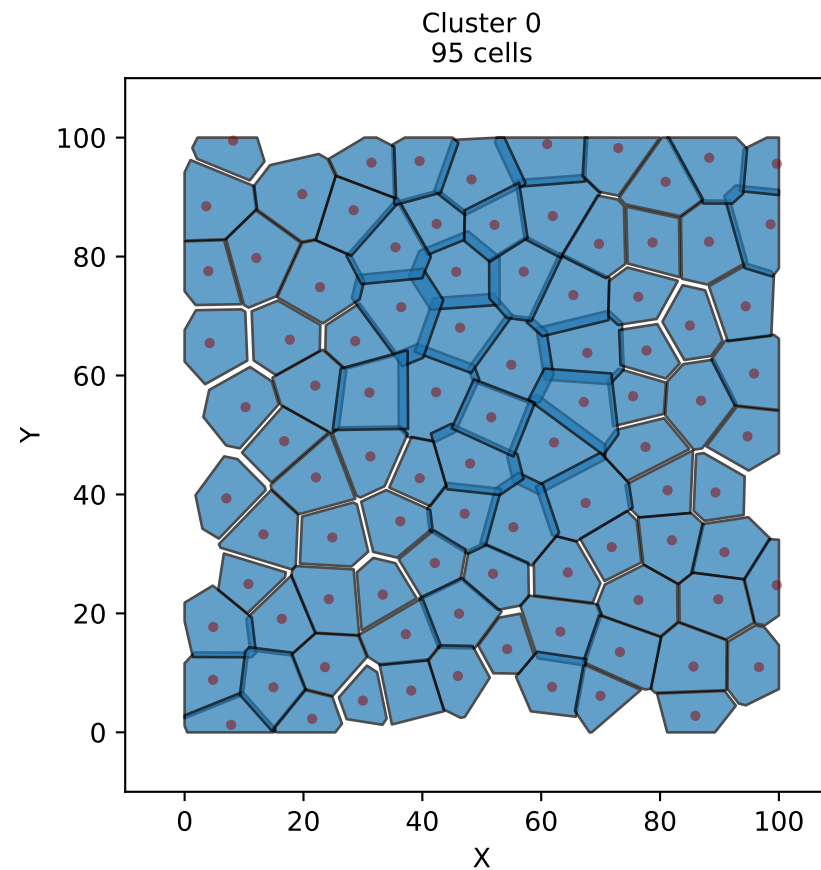


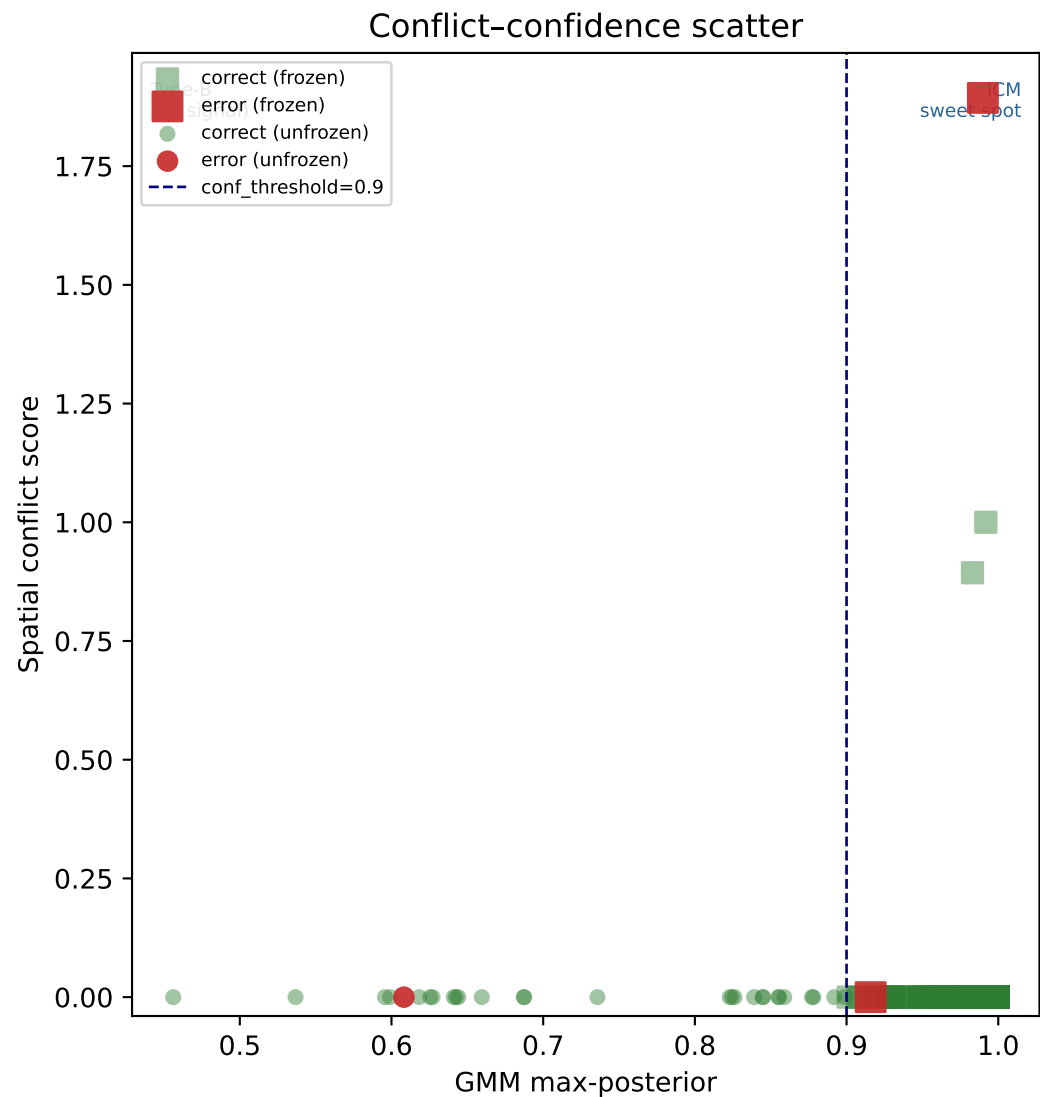
elongated — ground truth



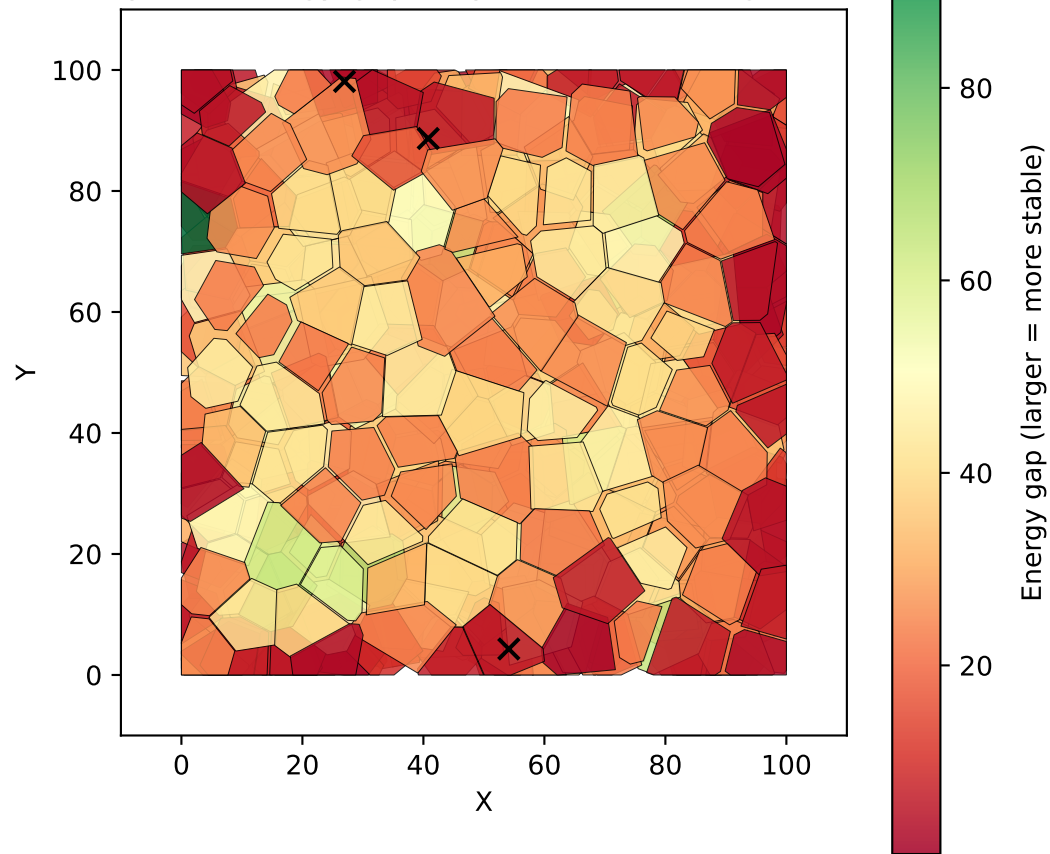
elongated — GMM + MRF ARI=0.985 violations 17 -> 2 changed=9 iters=2



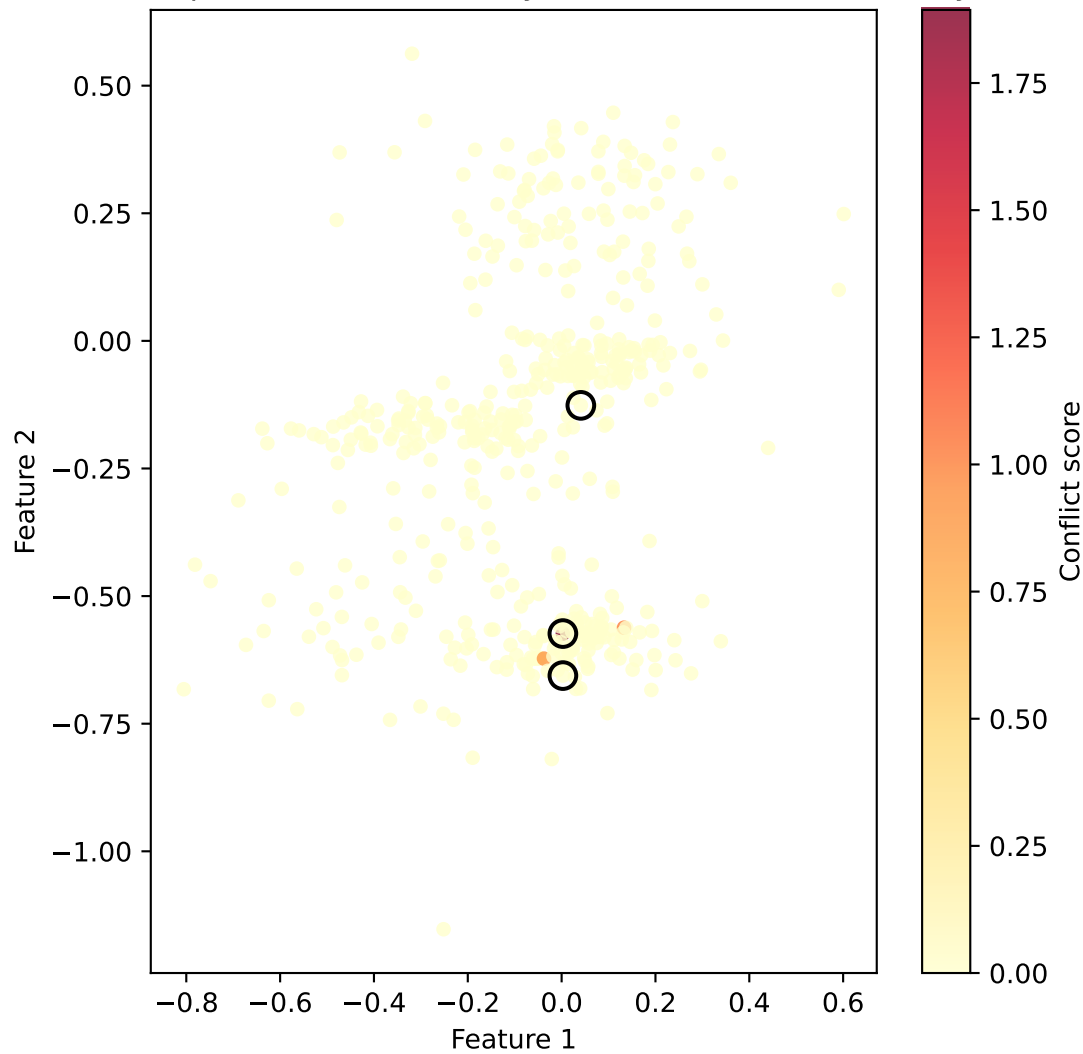
elongated — MRF diagnostics



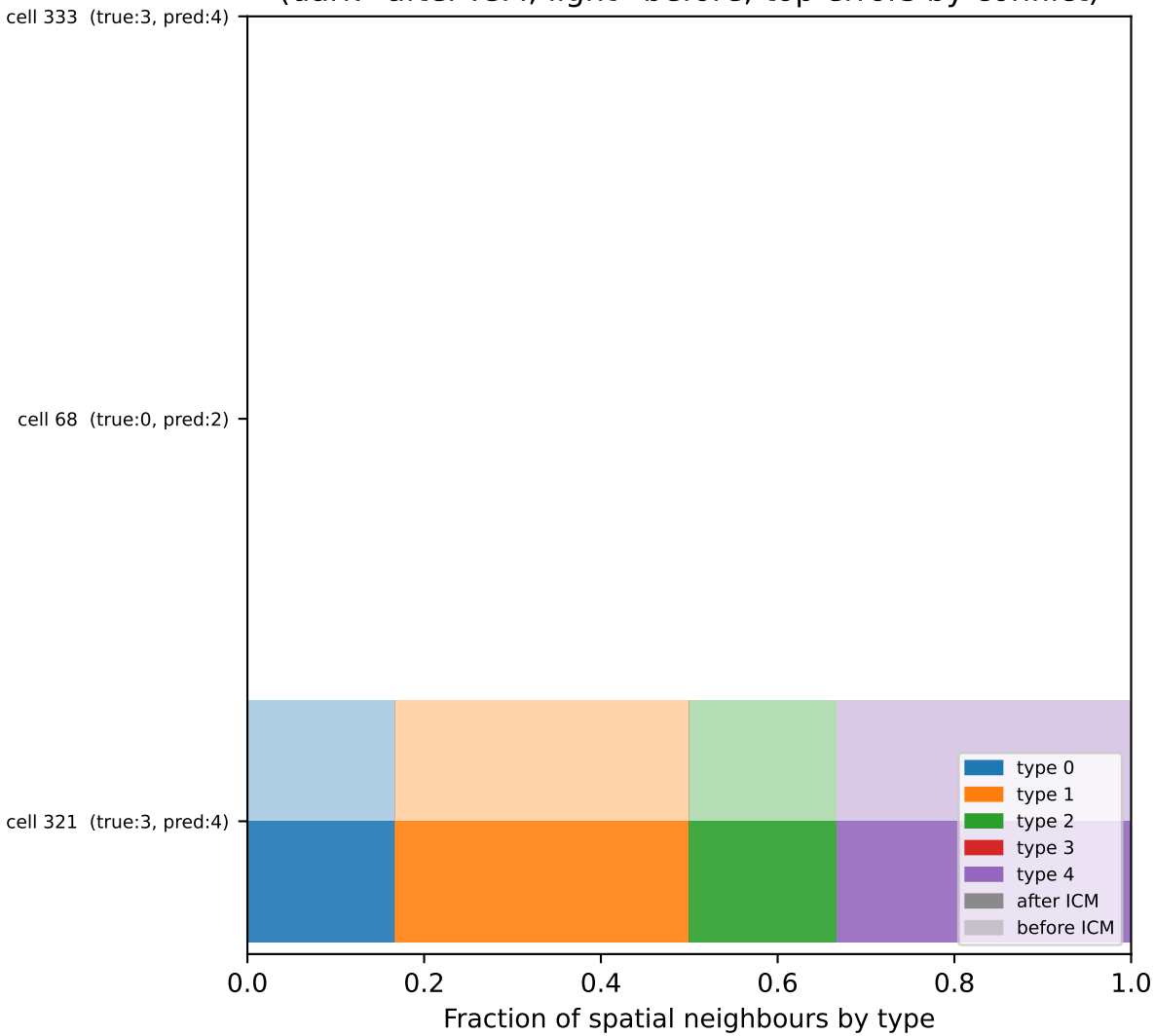
Spatial energy-gap map (x = remaining error)



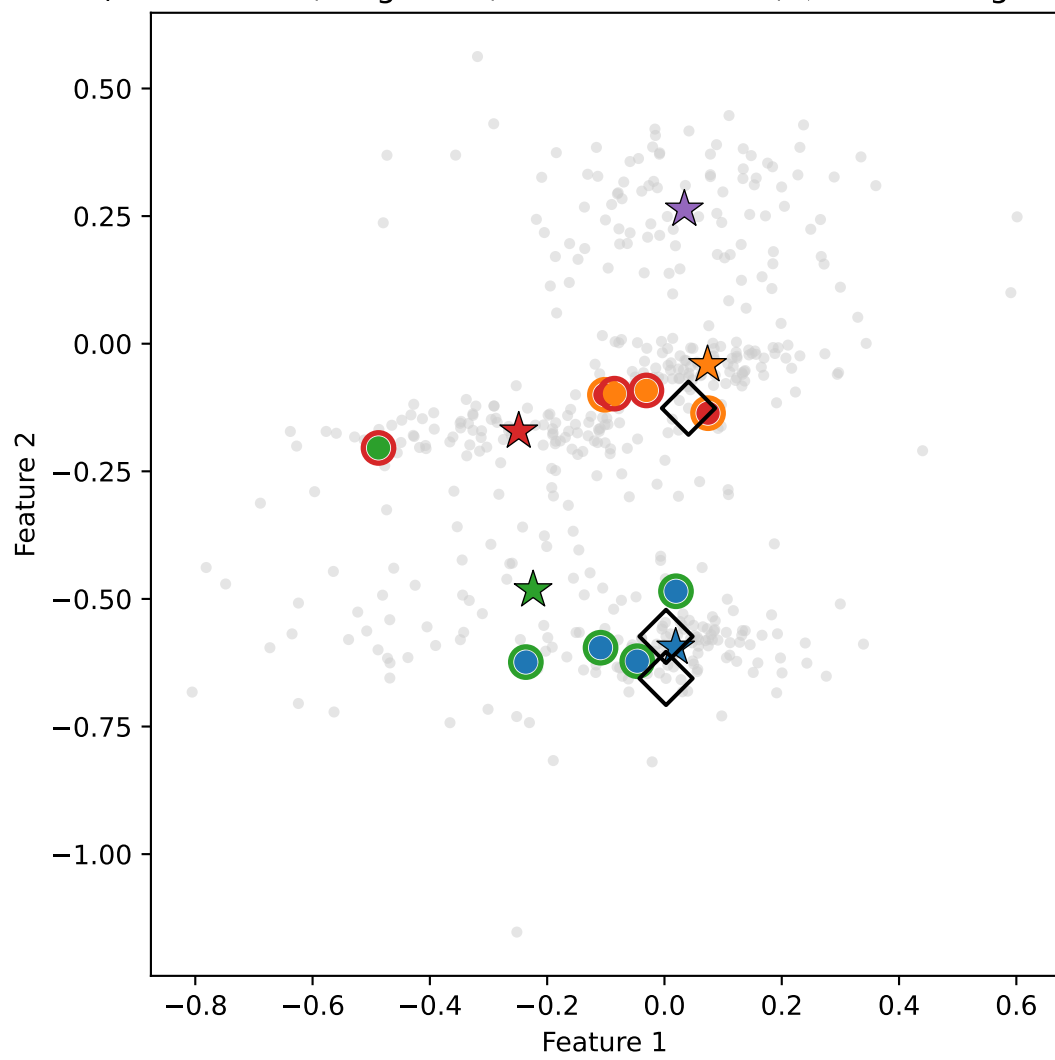
Feature space + conflict (navy = GMM decision boundary)



Neighbour type composition
(dark=after ICM, light=before; top errors by conflict)



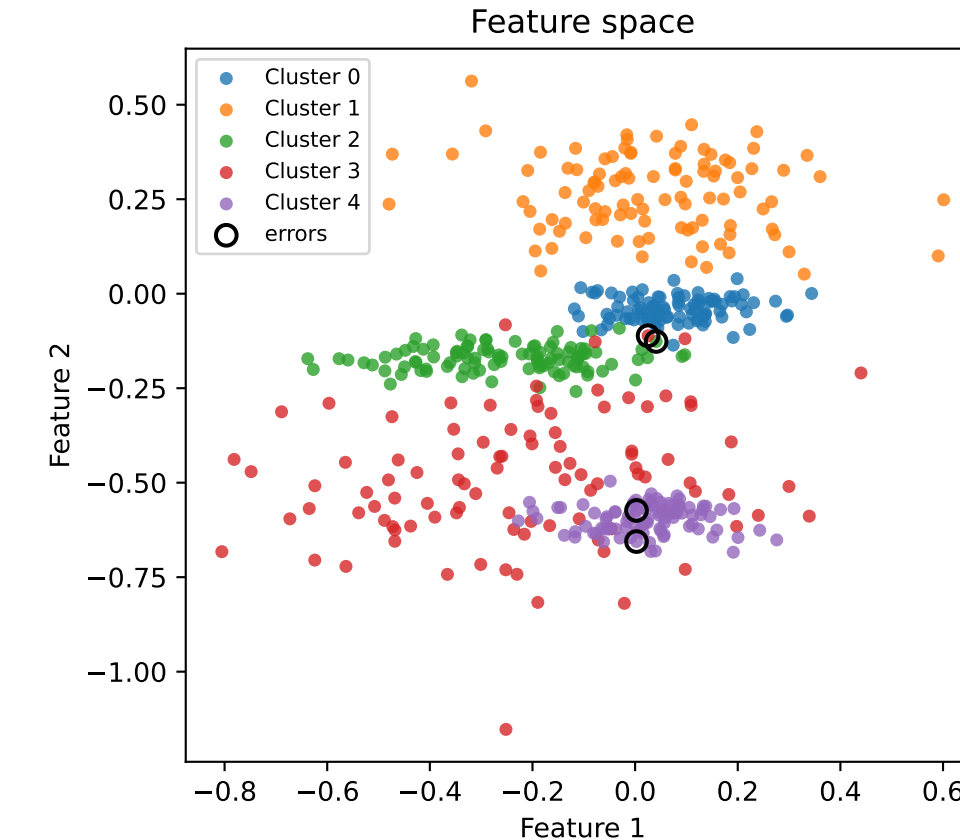
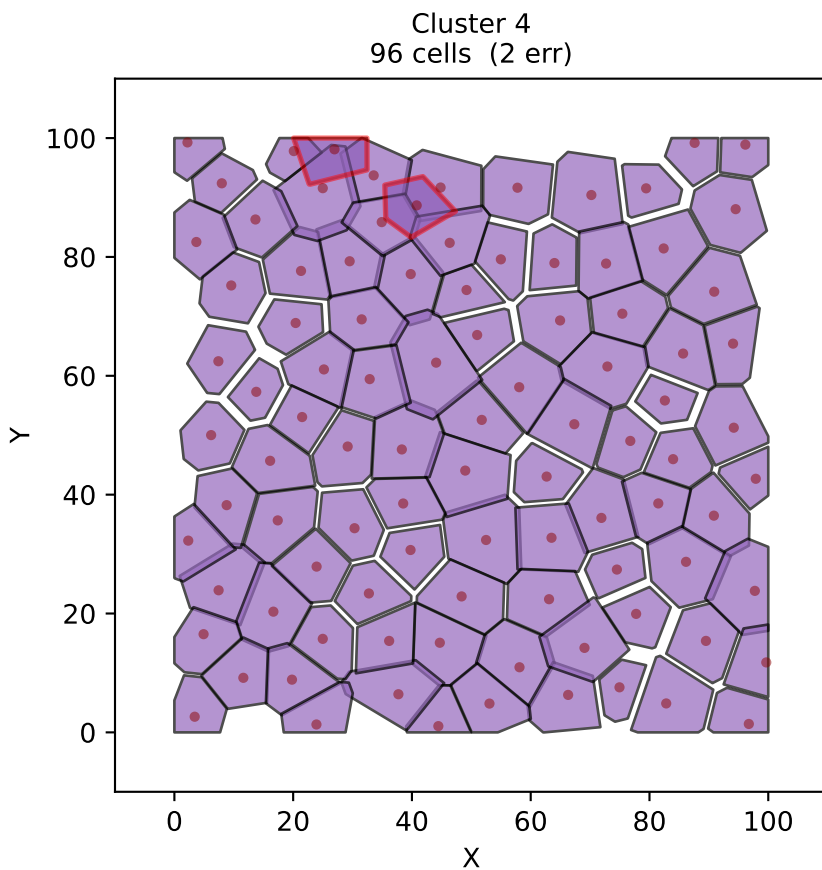
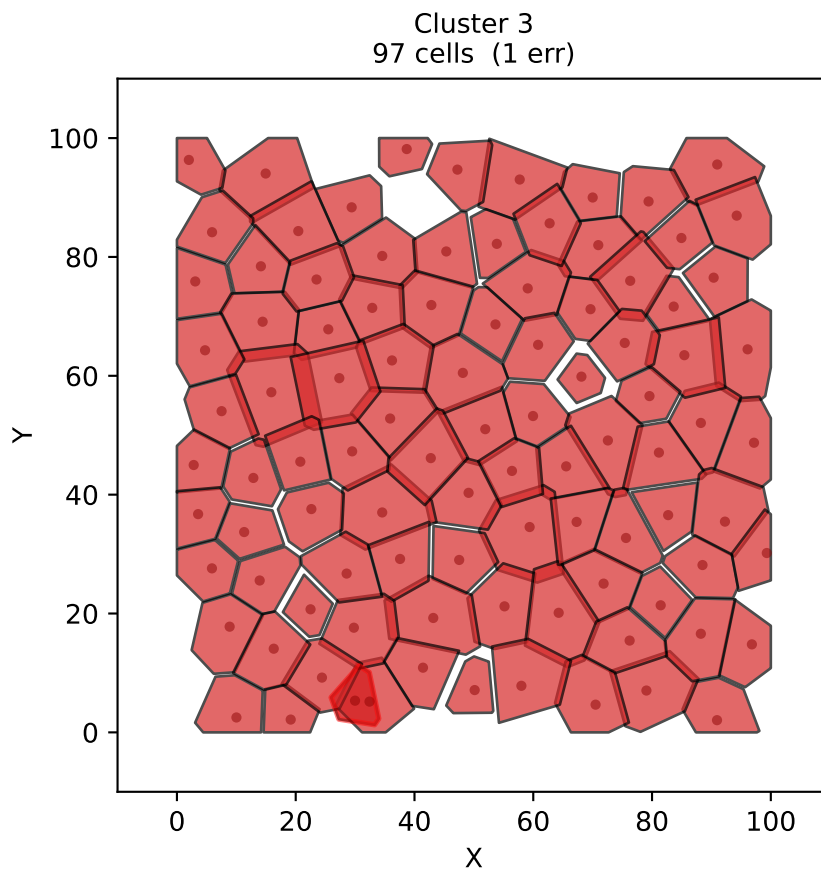
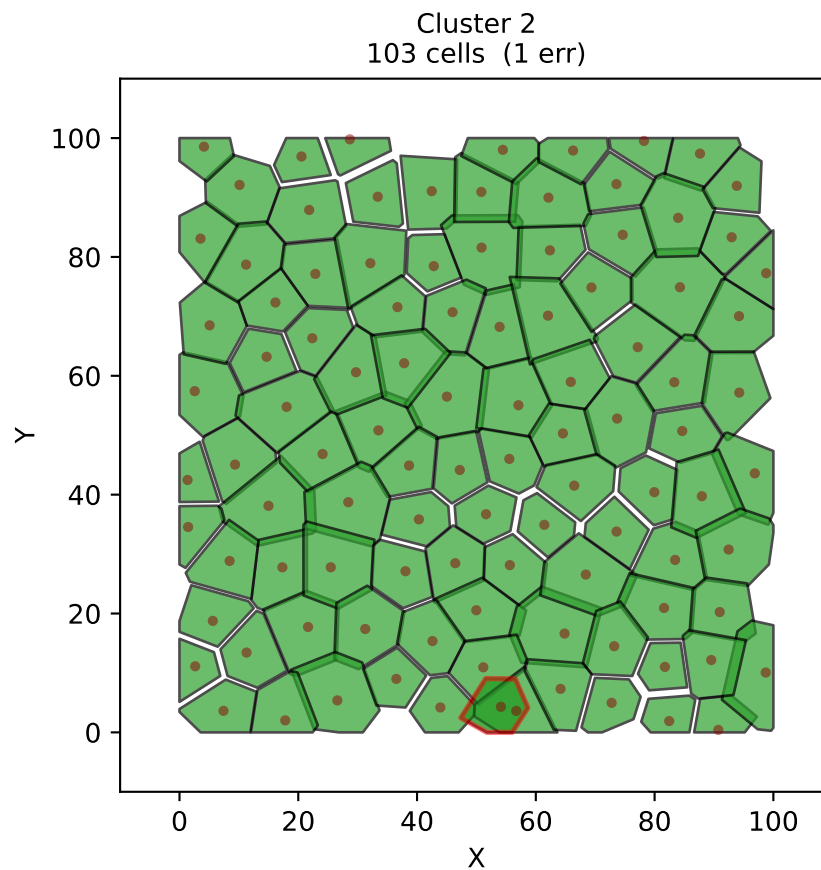
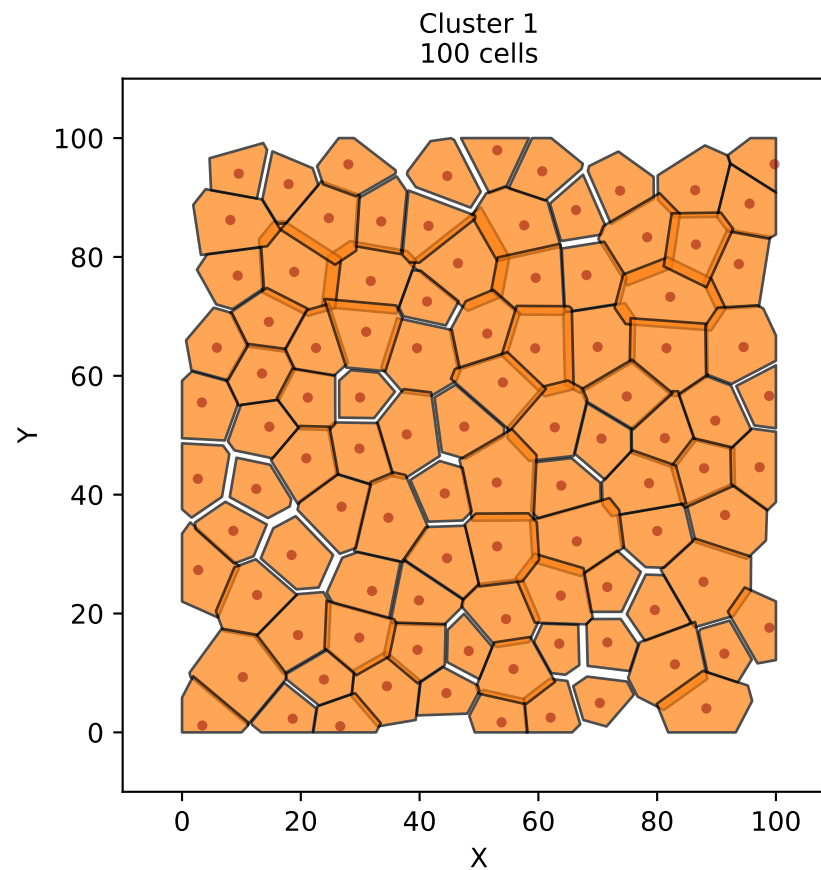
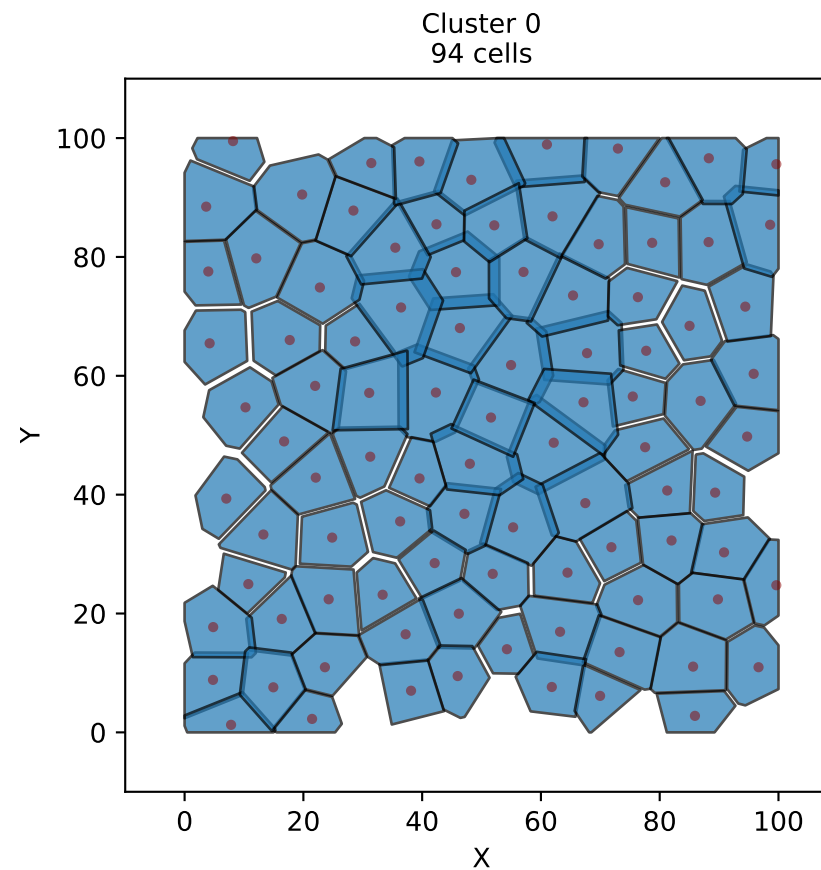
ICM label changes: 9 cells moved
(fill=old label, ring=new; ★=cluster mean; ◇=remaining error)

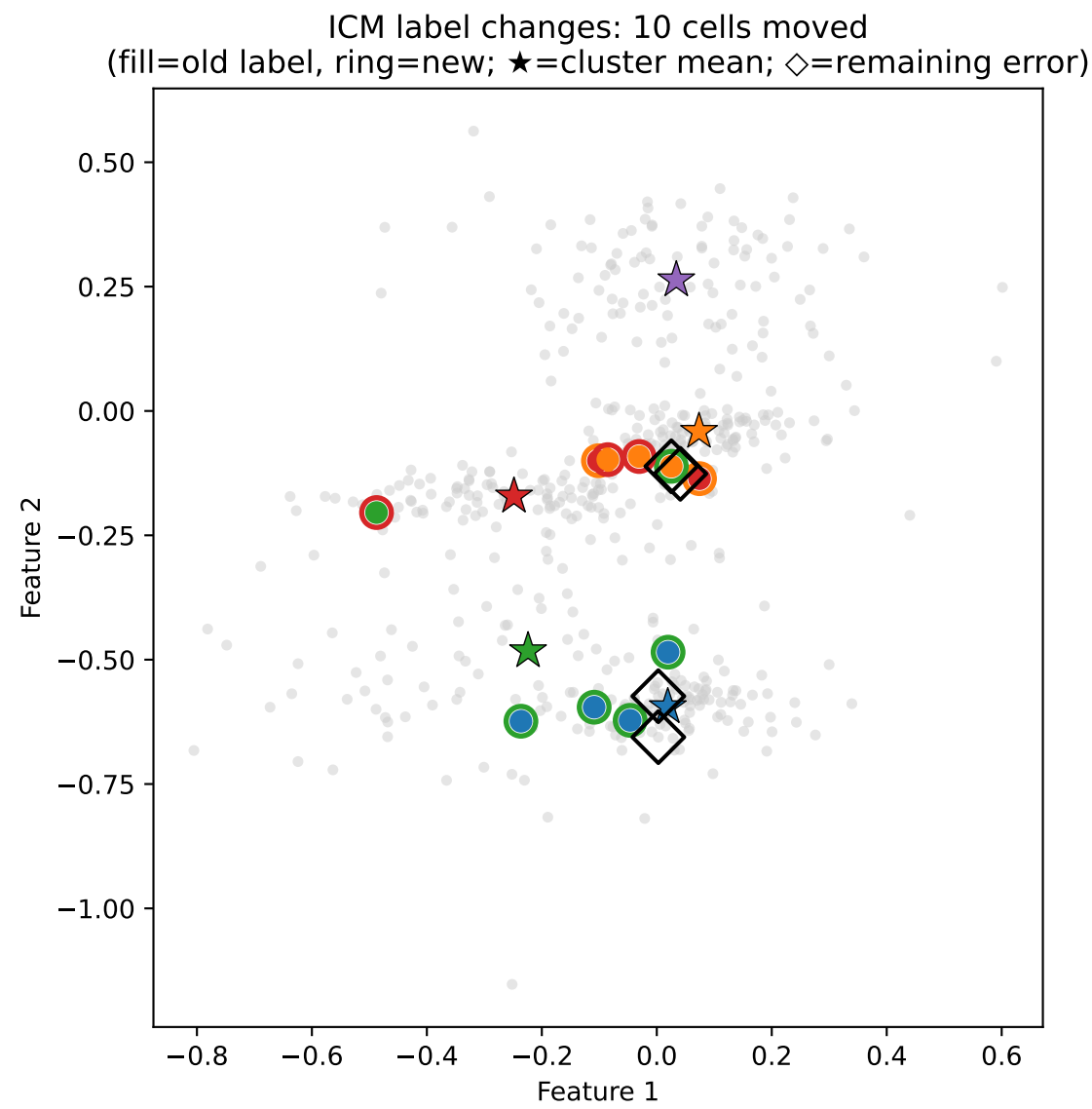
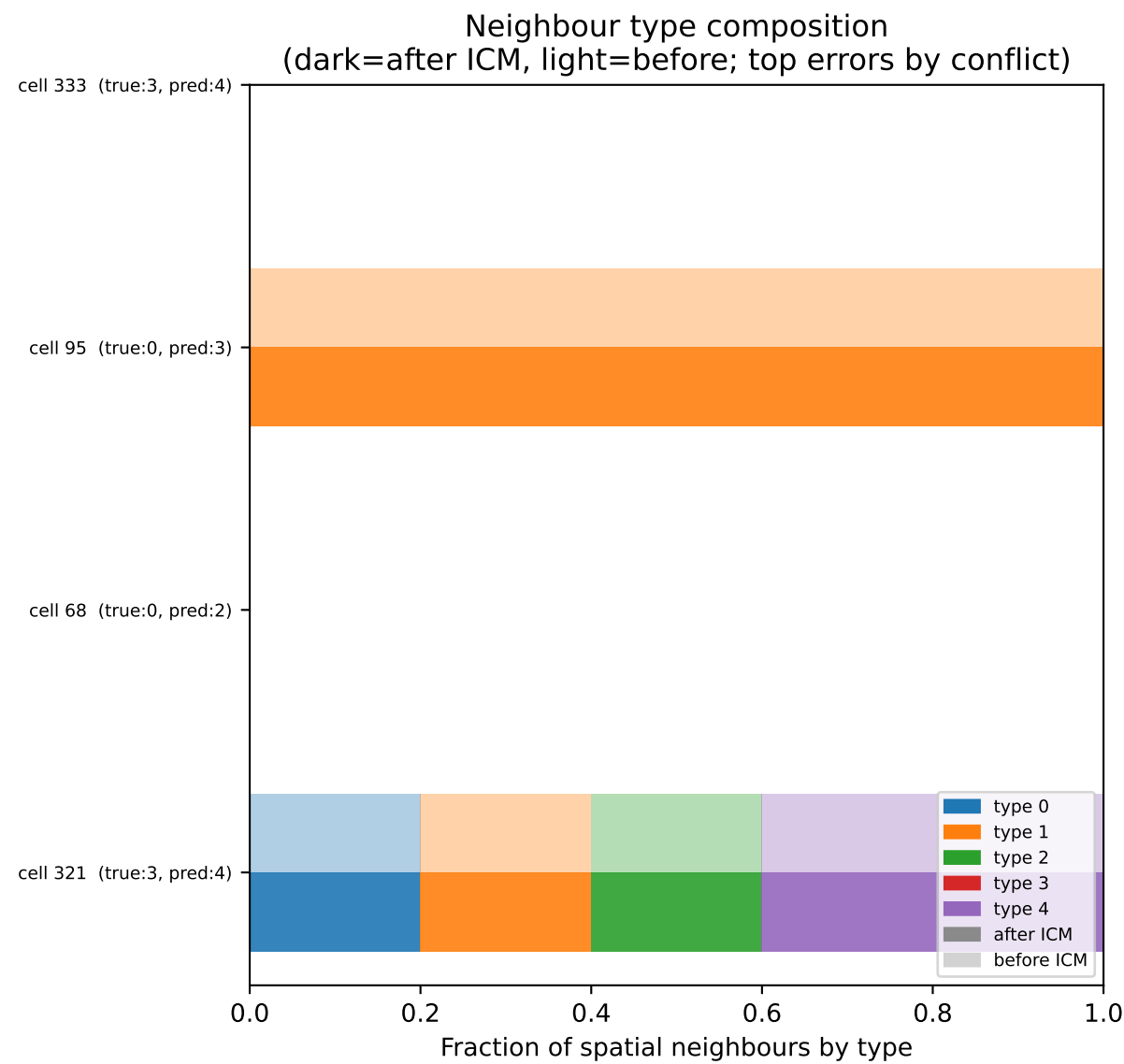
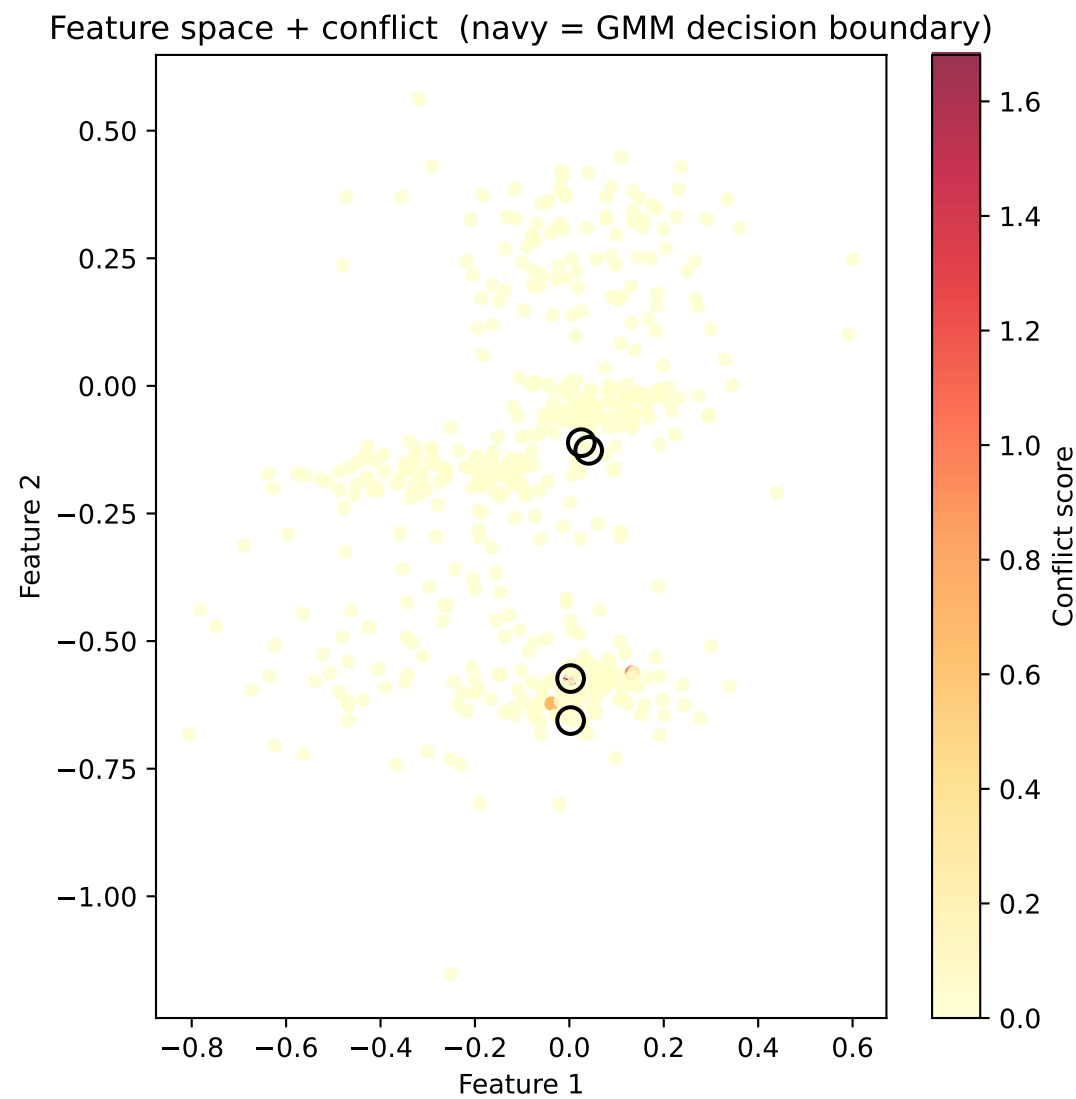
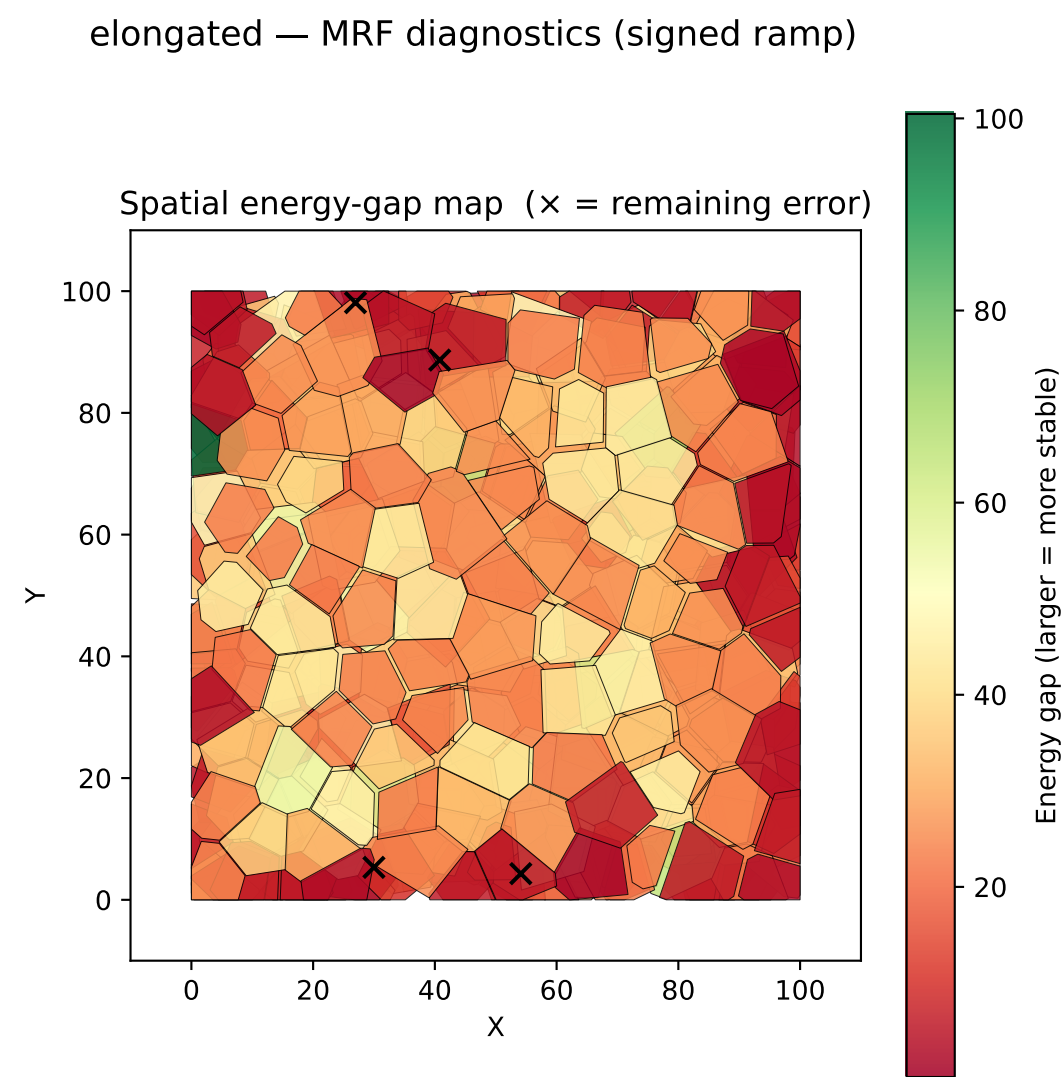
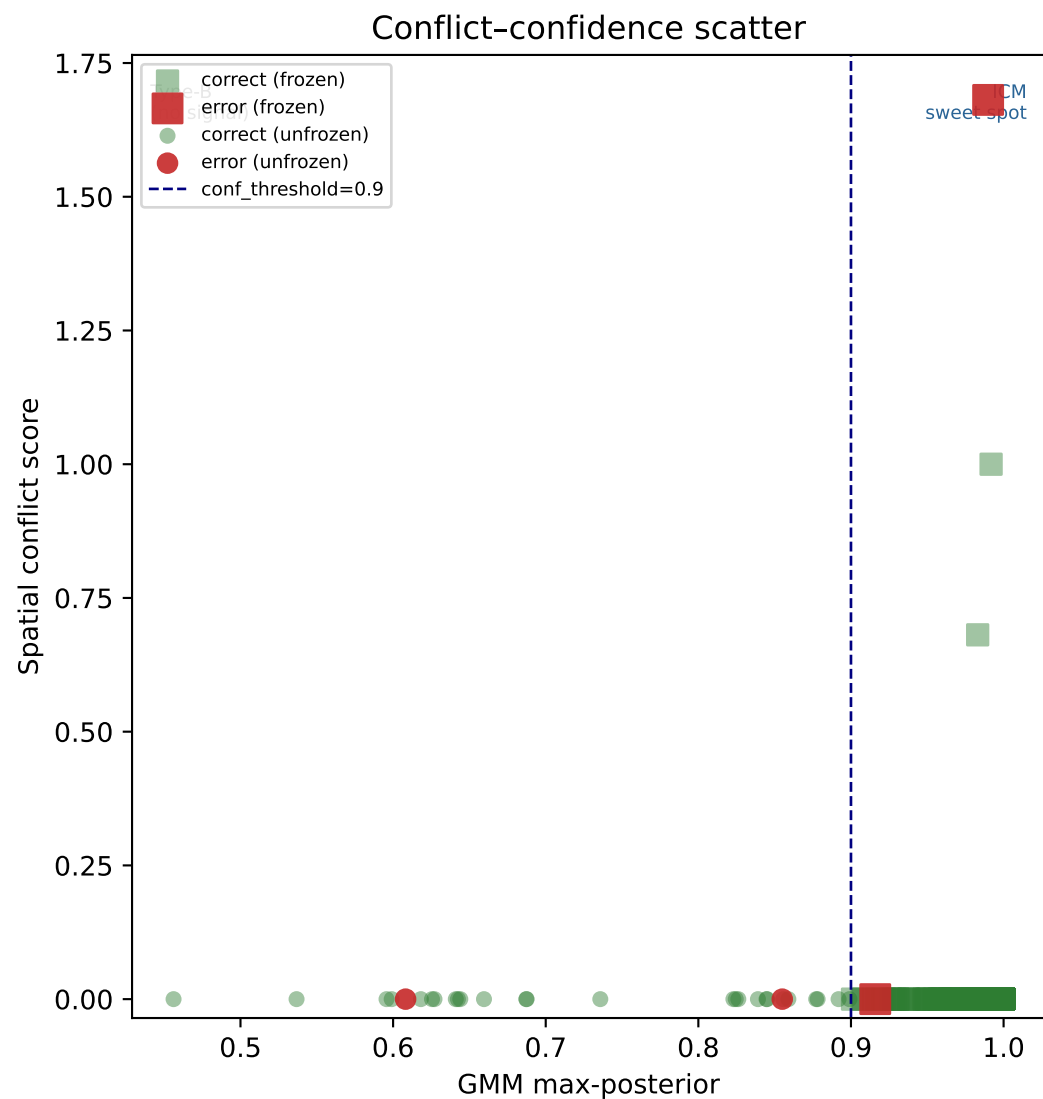


Run summary

MRF Run Summary	
ARI:	0.9849
errors:	3 / 490
violations:	17 → 2
n_changed:	9
n_iters:	2
n_frozen:	462
Parameters:	
lam:	20.0
tau_low:	0.1
tau_high:	0.4
radius:	7.059097824976937
conf_thr:	0.9

elongated — GMM + MRF (signed ramp, tau_low=0.3) ARI=0.980 violations 16 -> 2 changed=10 iters=2

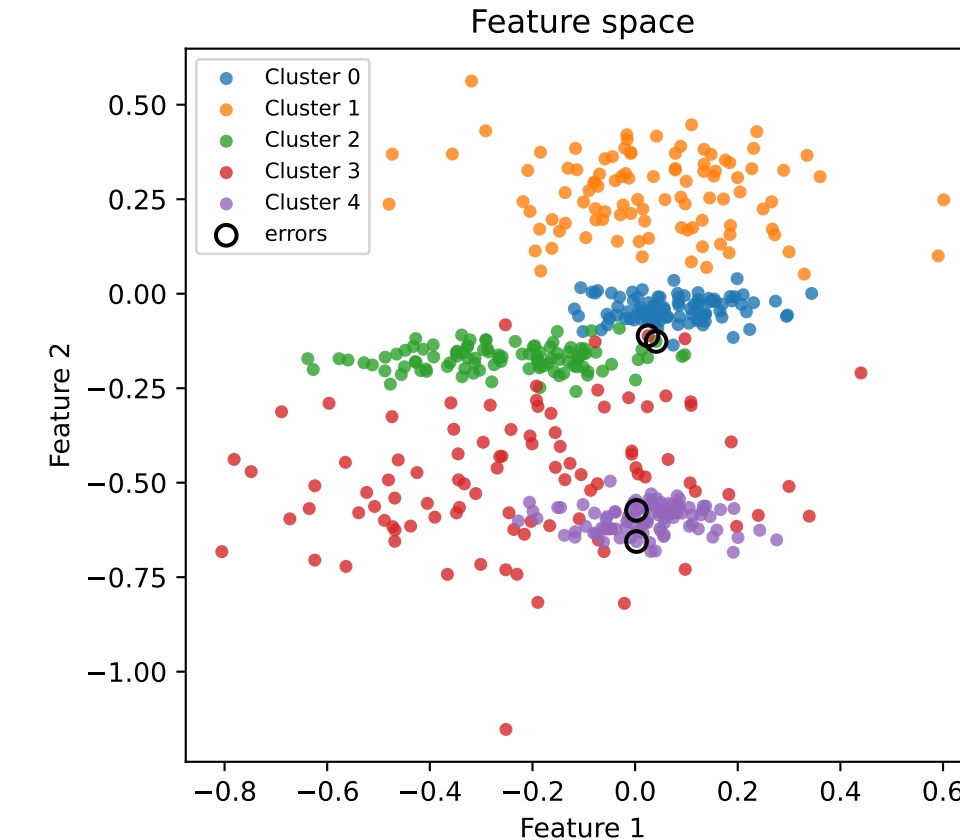
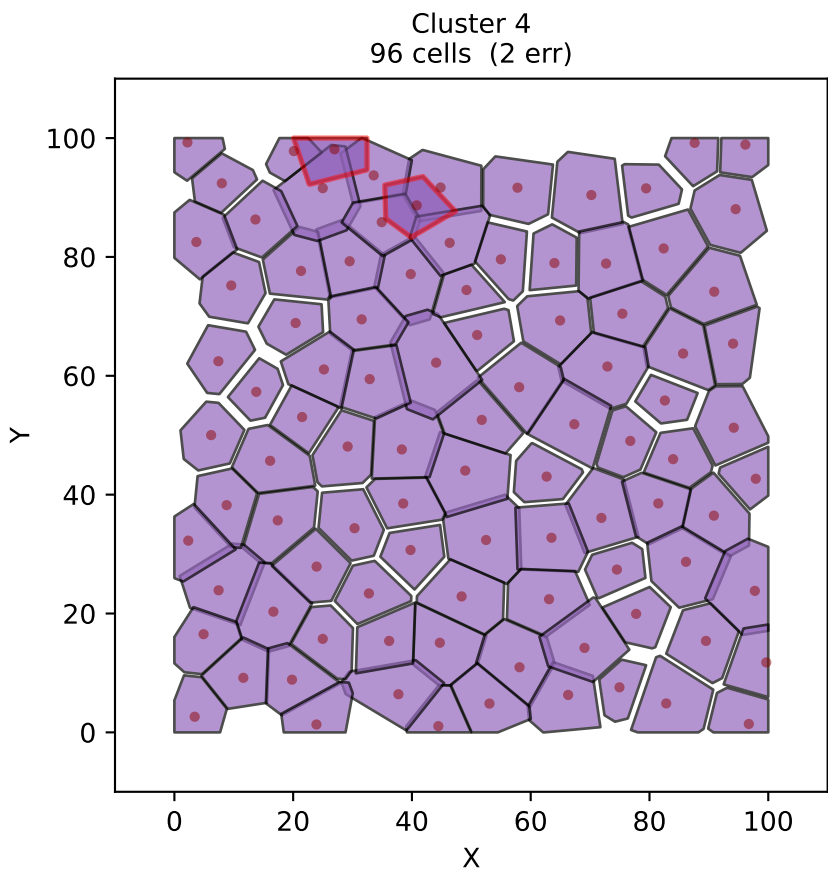
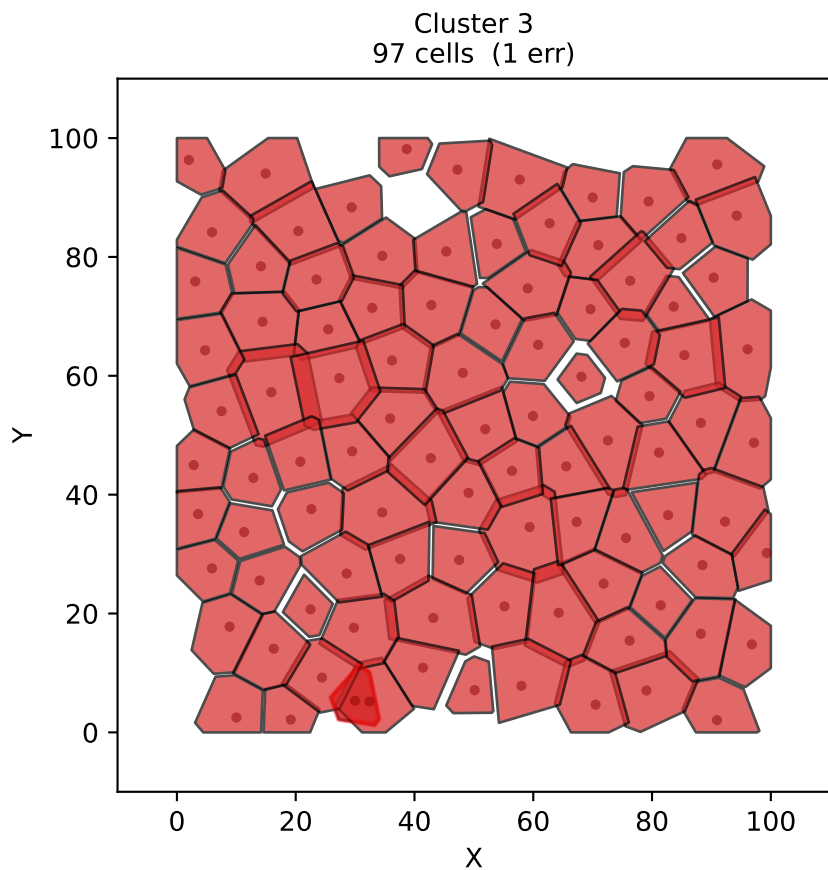
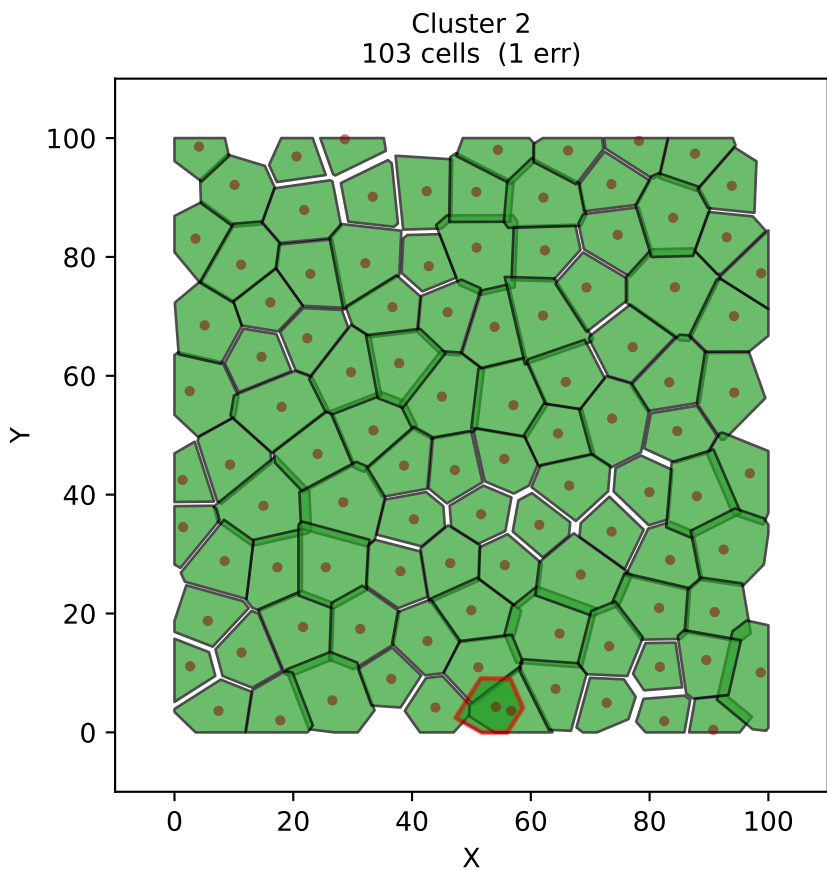
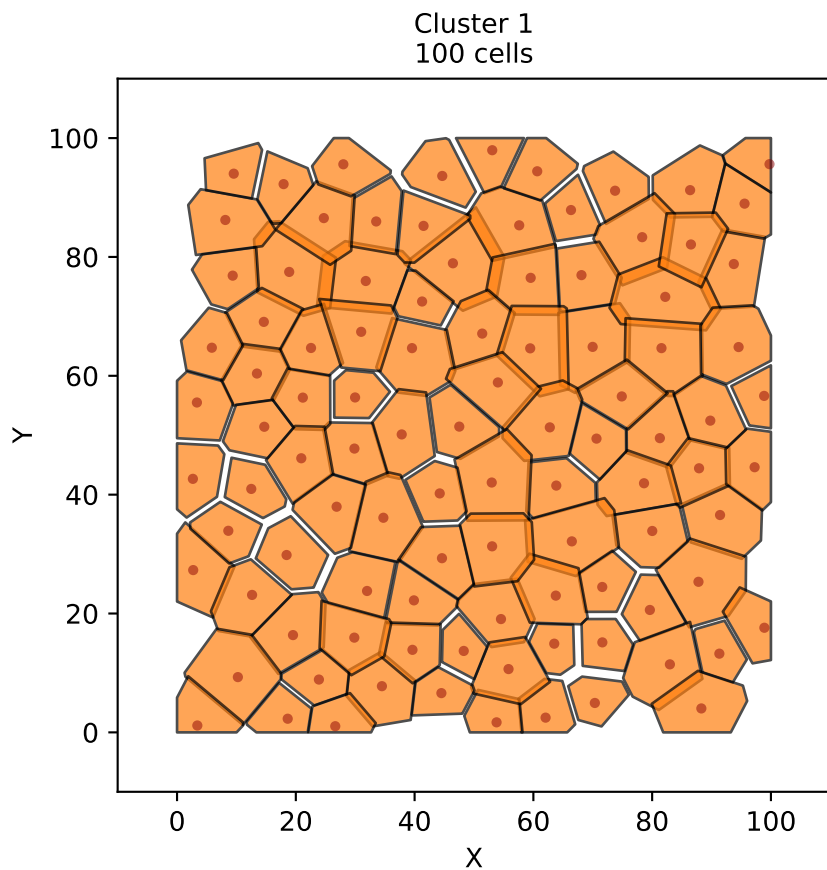
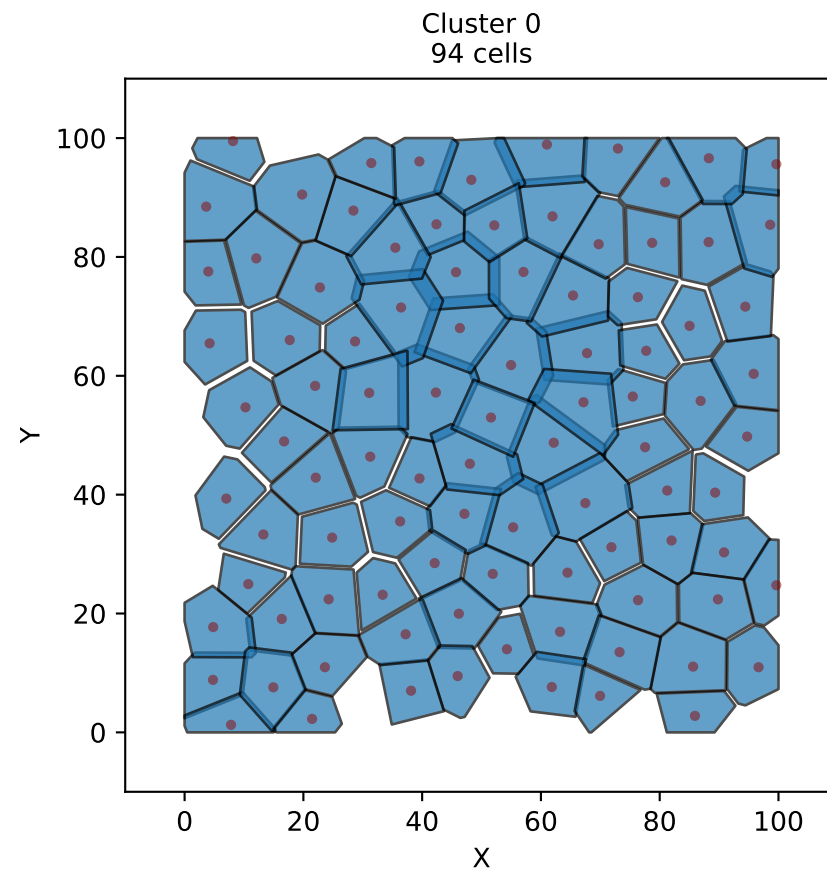




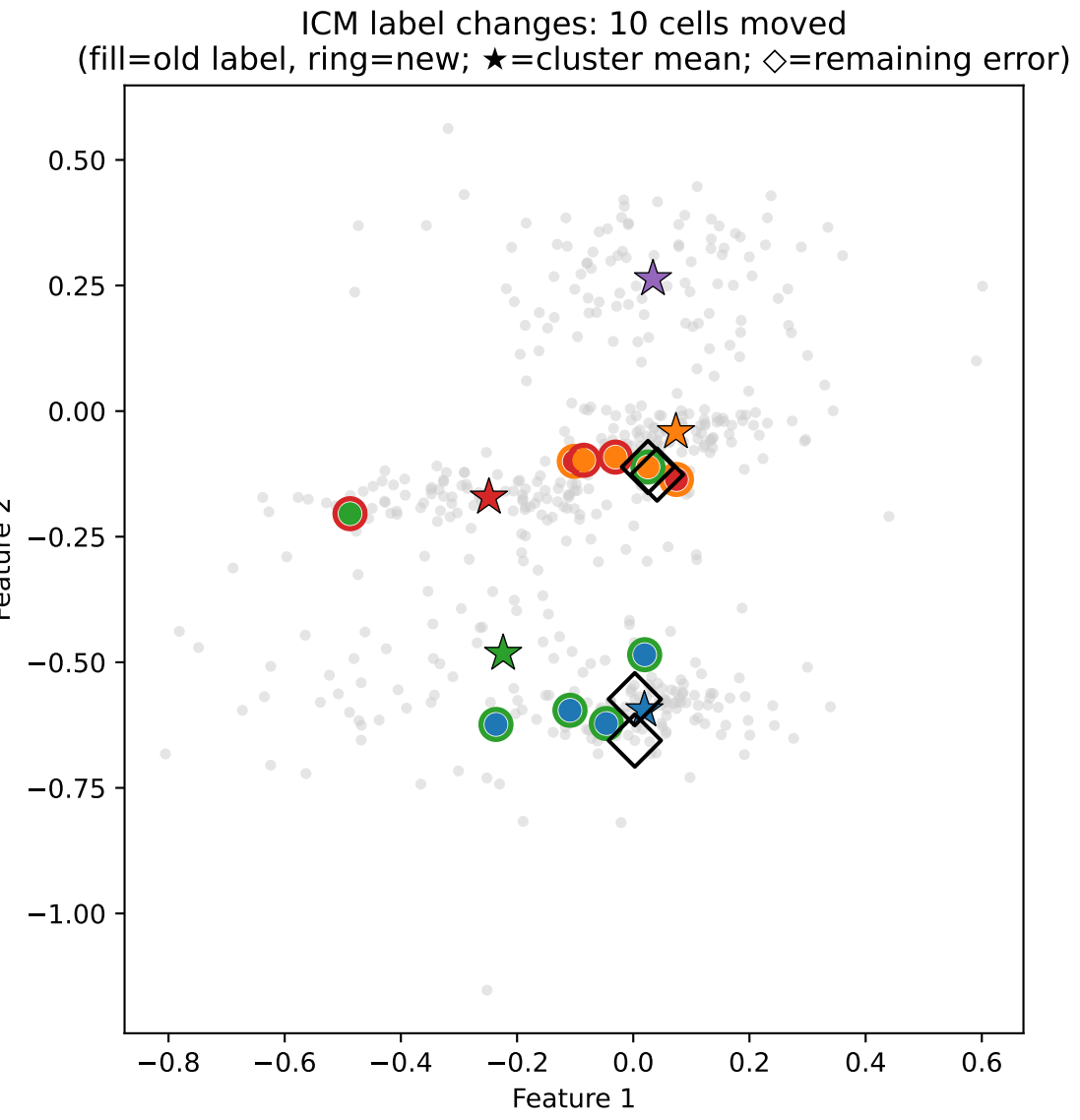
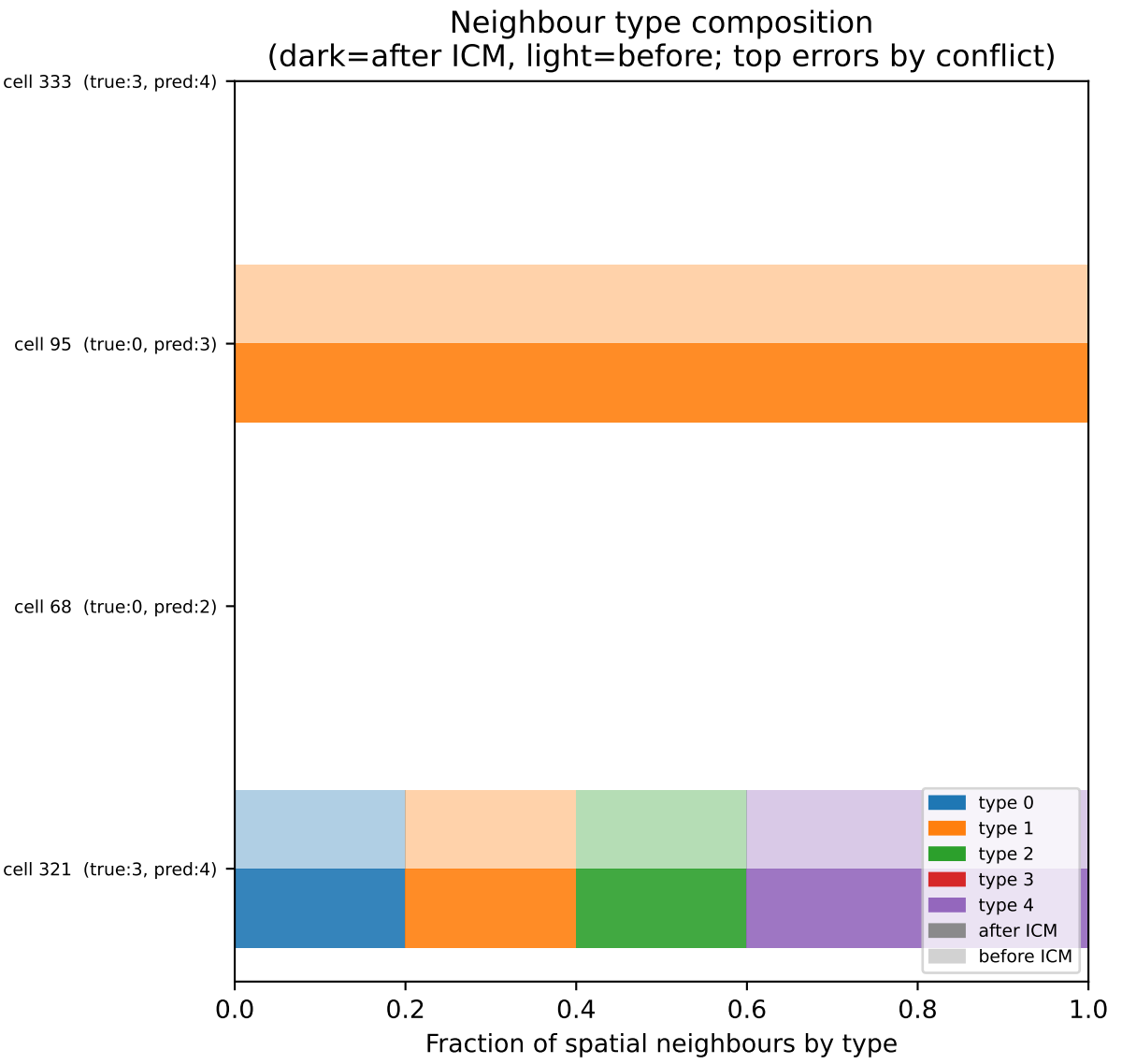
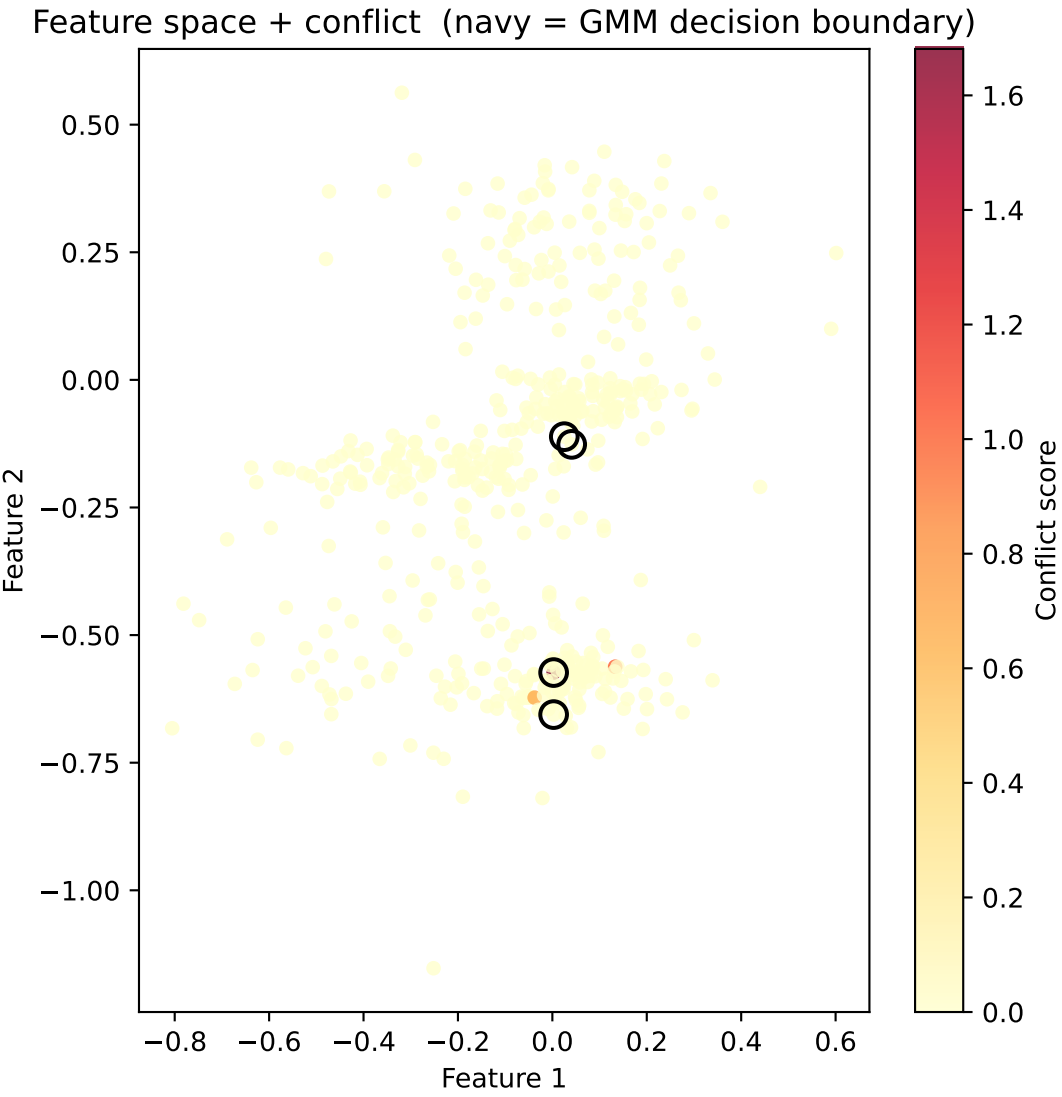
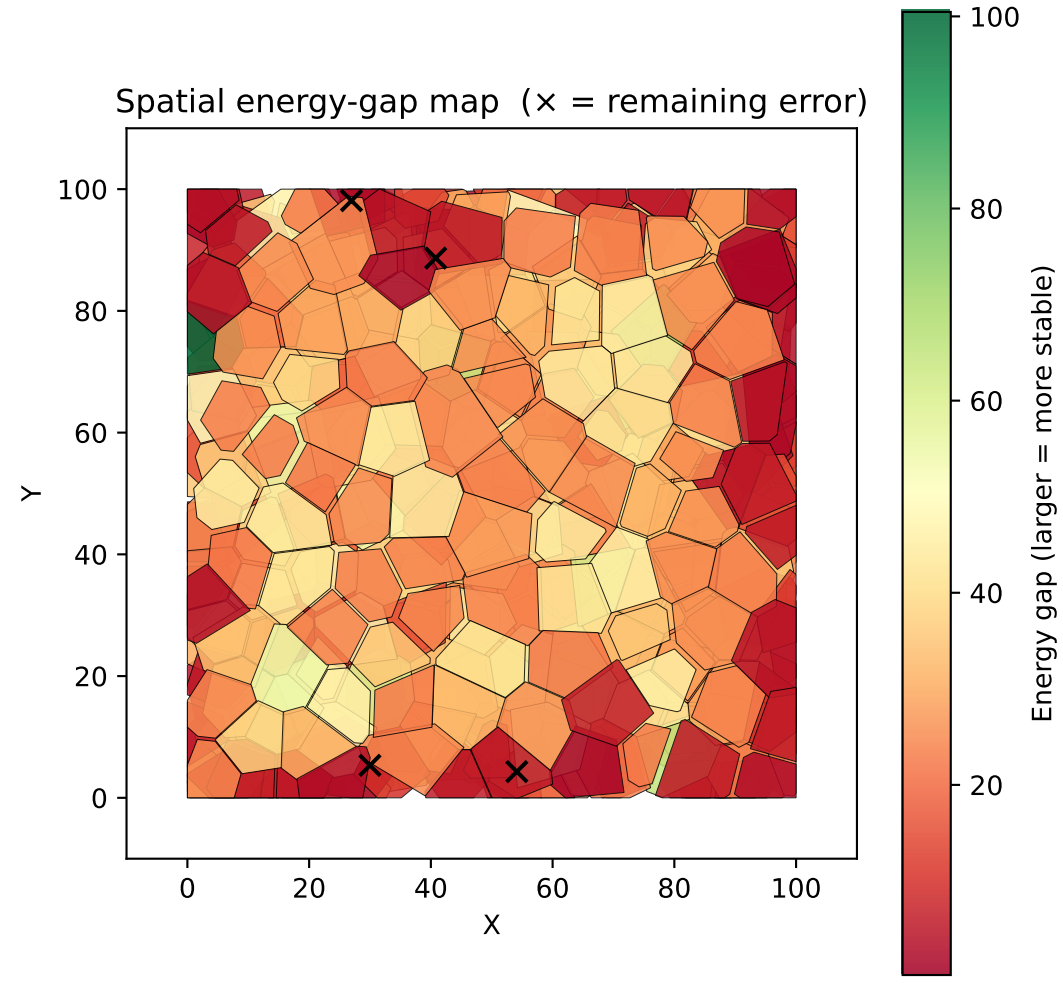
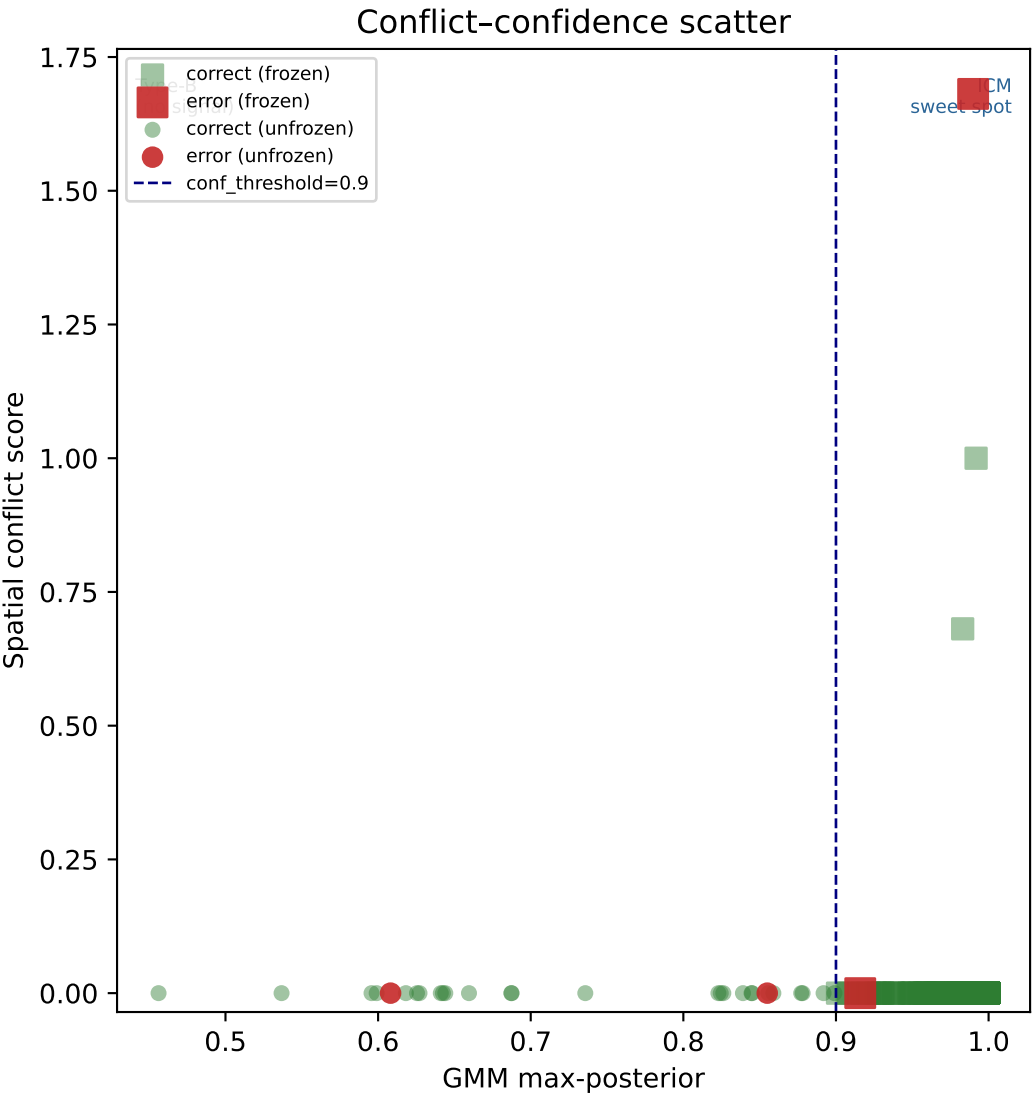
Run summary

MRF Run Summary	
ARI:	0.9799
errors:	4 / 490
violations:	16 → 2
n_changed:	10
n_iters:	2
n_frozen:	462
Parameters:	
lam:	20.0
tau_low:	0.3
tau_high:	0.4
radius:	7.059097824976937
conf_thr:	0.9

elongated — GMM + MRF (signed log-ramp $\alpha=10$, $\tau_{\text{low}}=0.3$) ARI=0.980 violations 16 -> 2 changed=10 iters=2



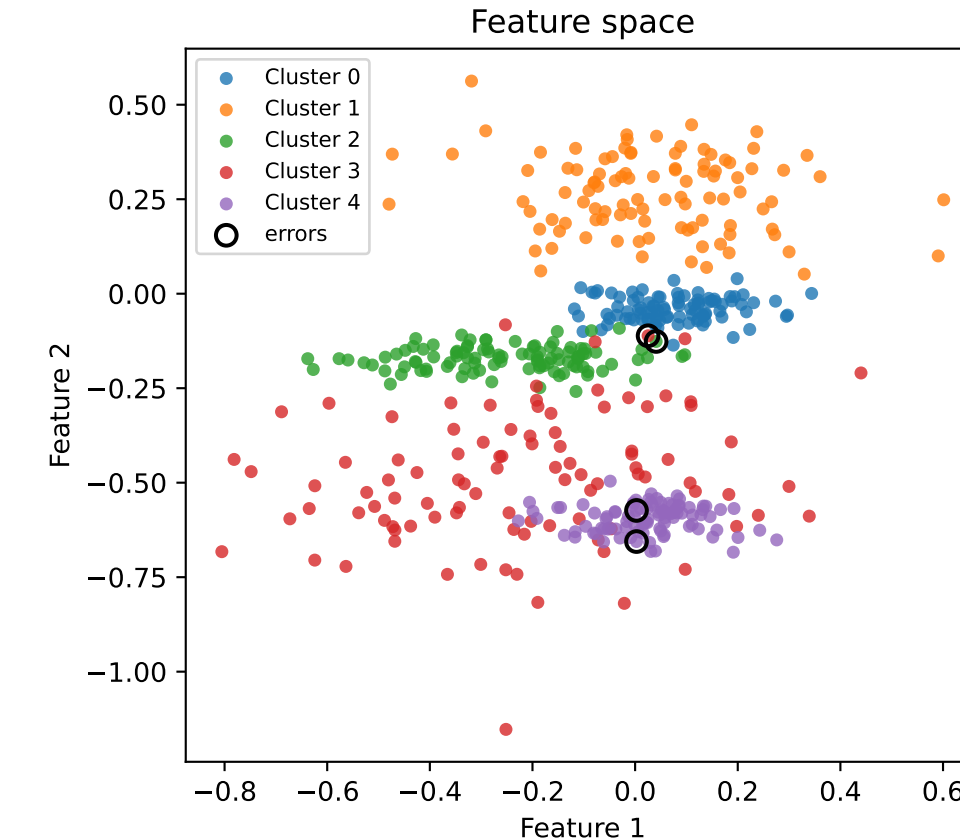
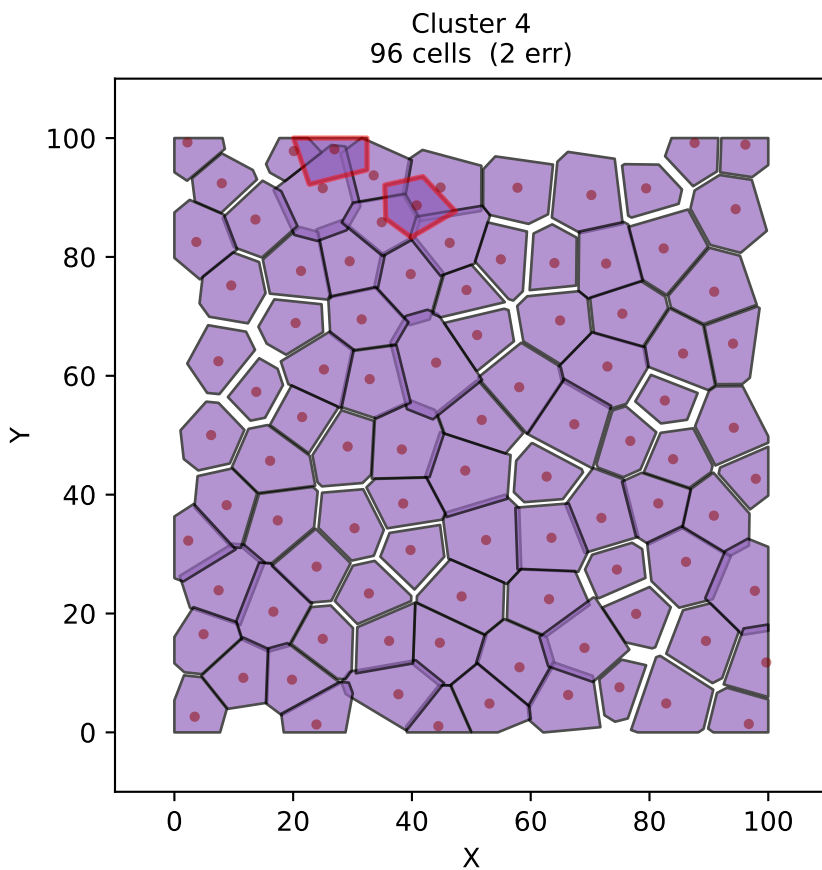
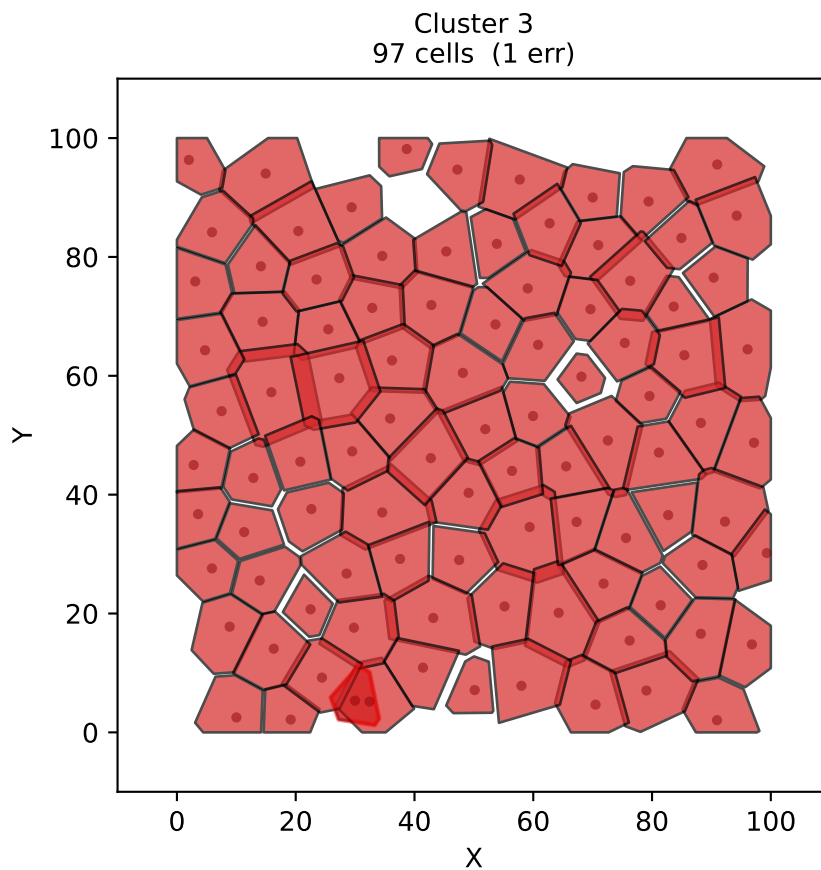
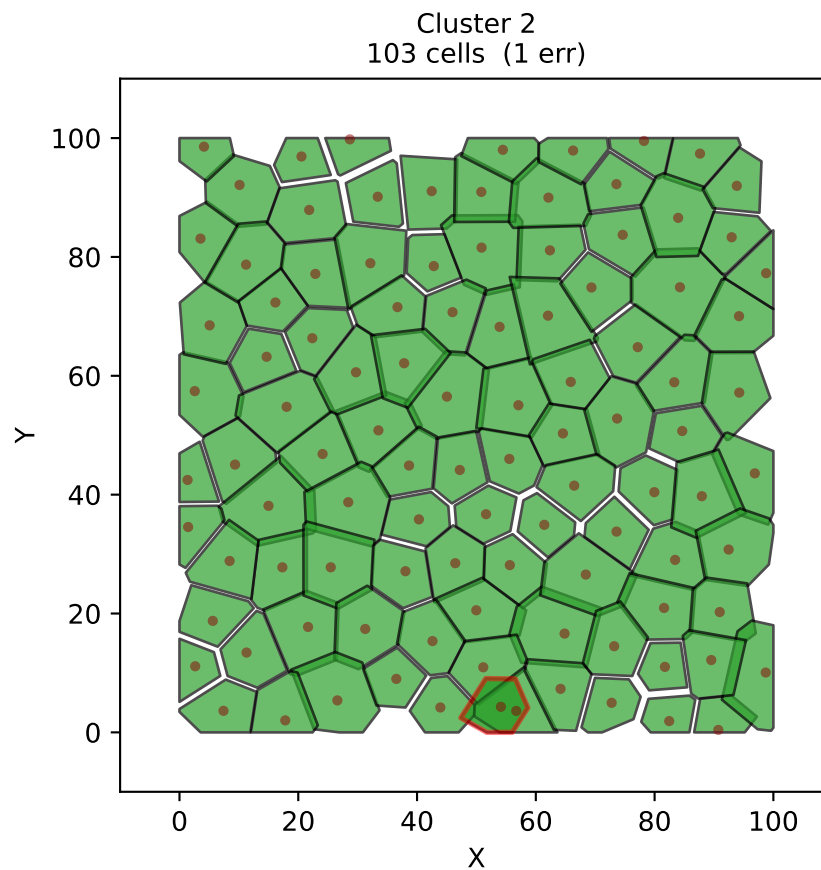
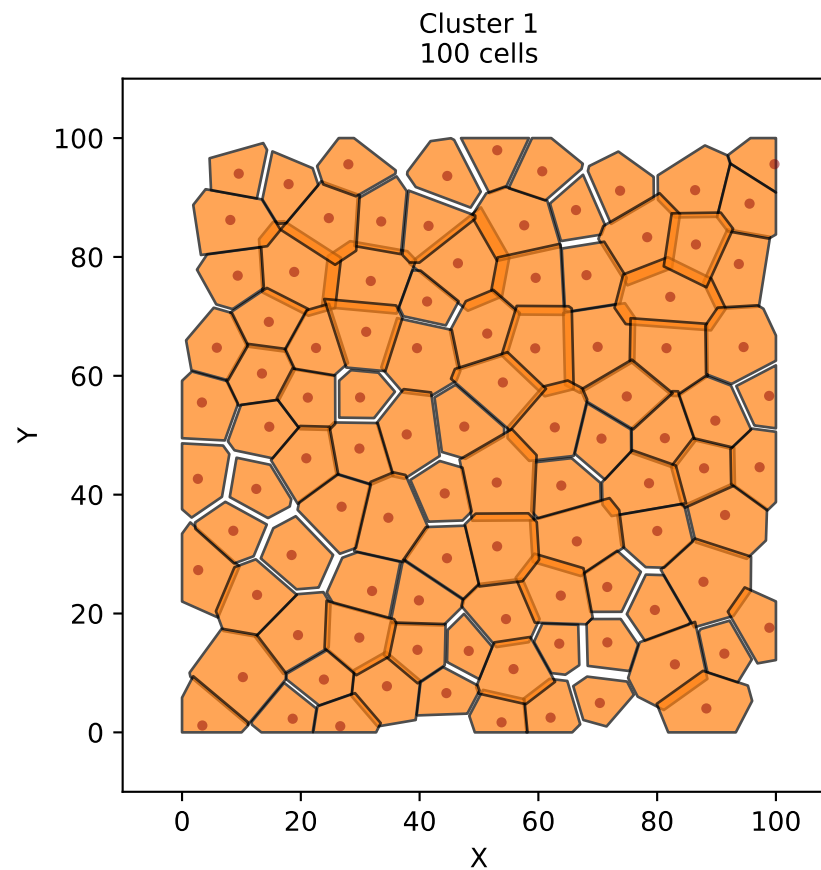
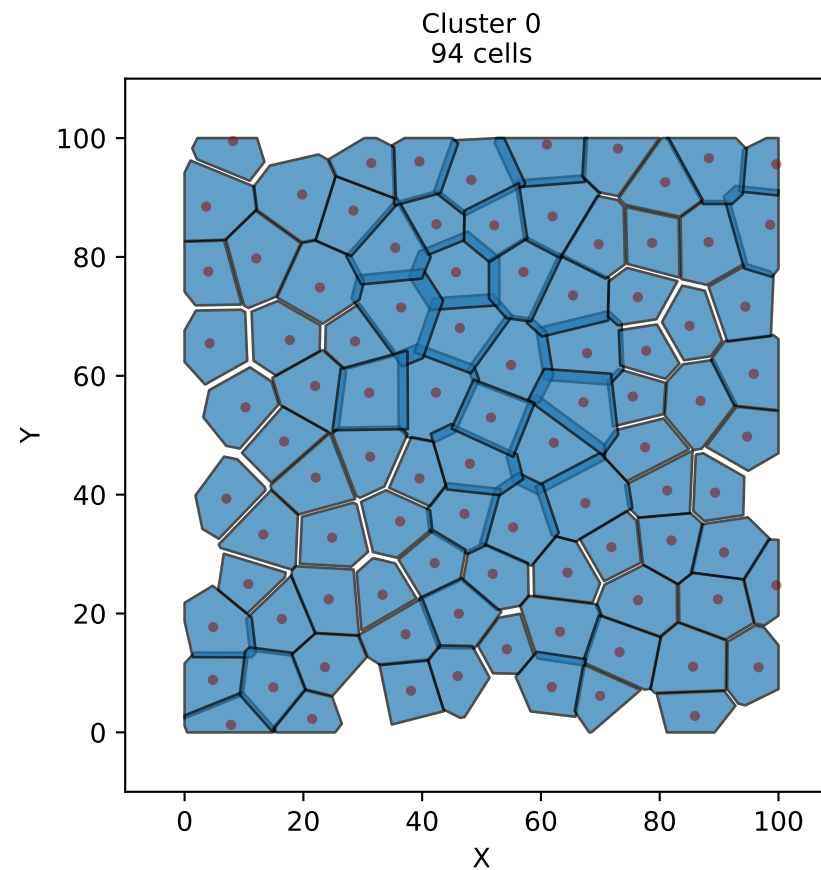
elongated — MRF diagnostics (signed log-ramp $\alpha=10$)



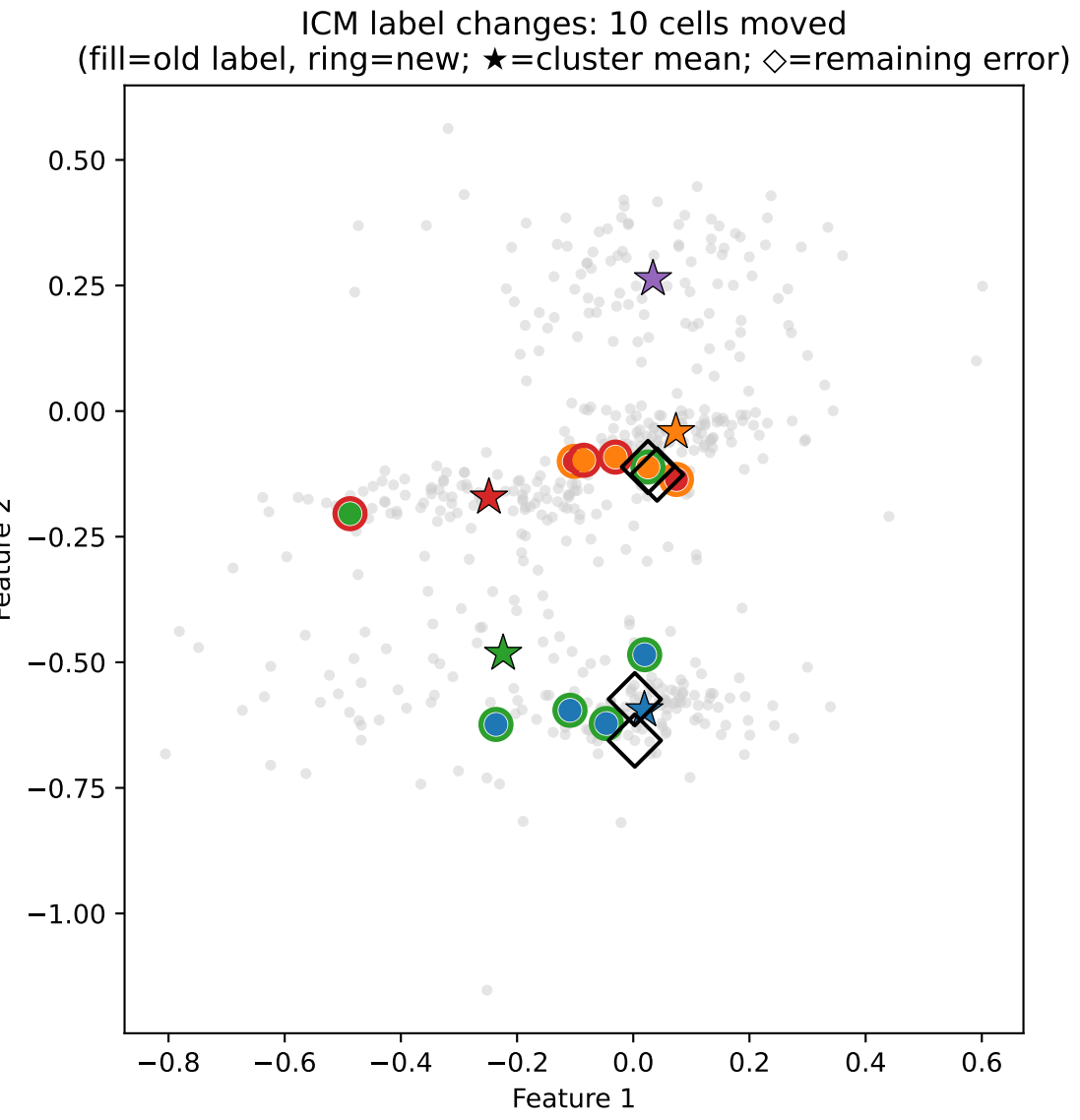
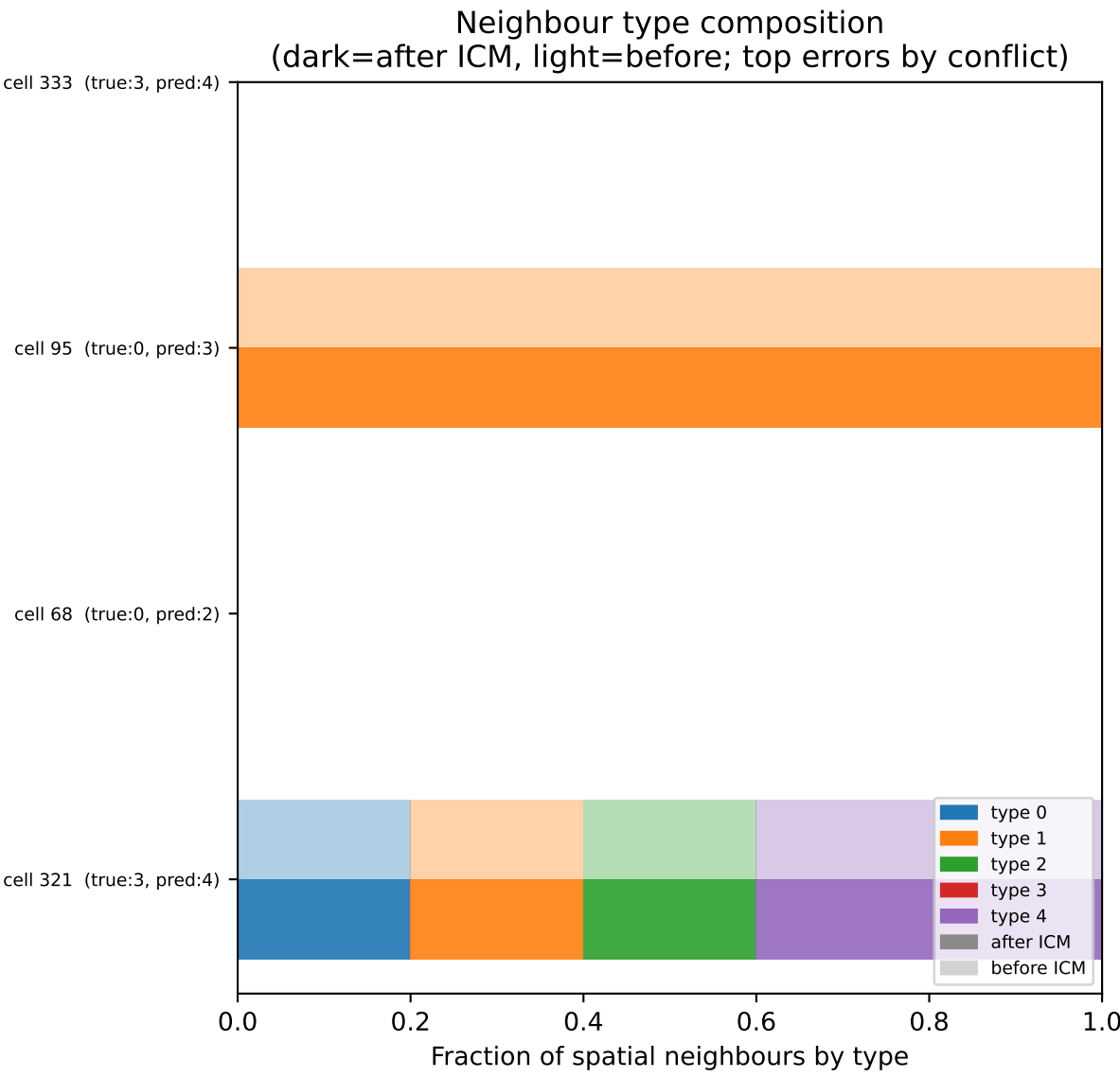
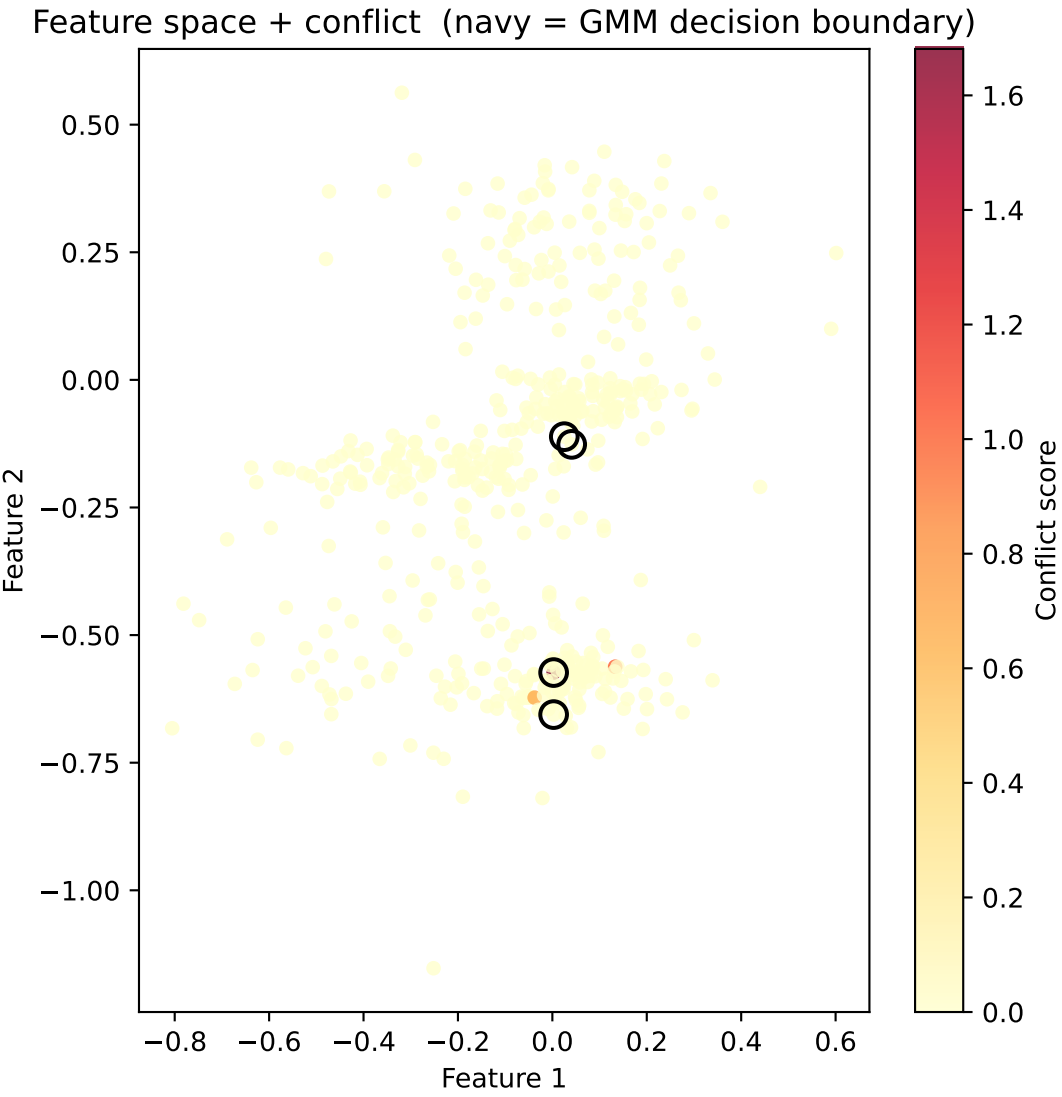
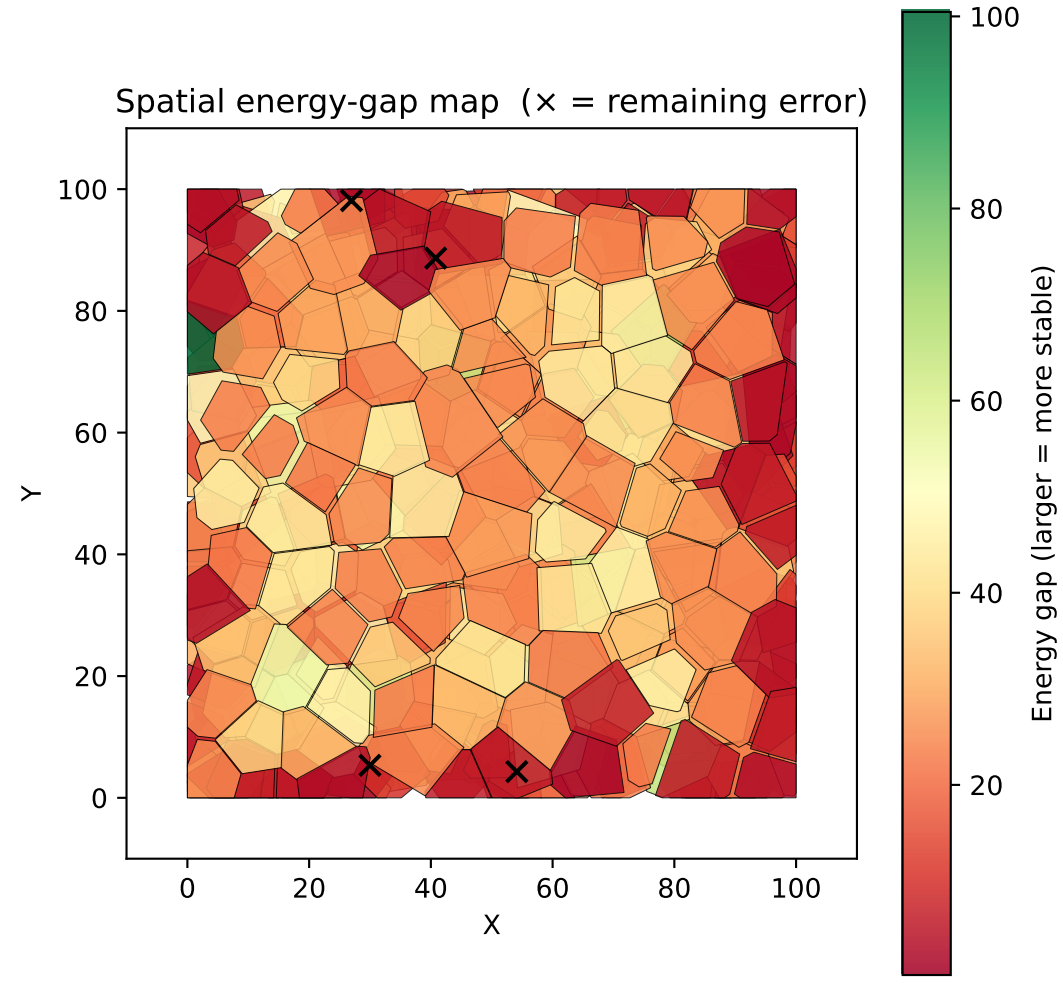
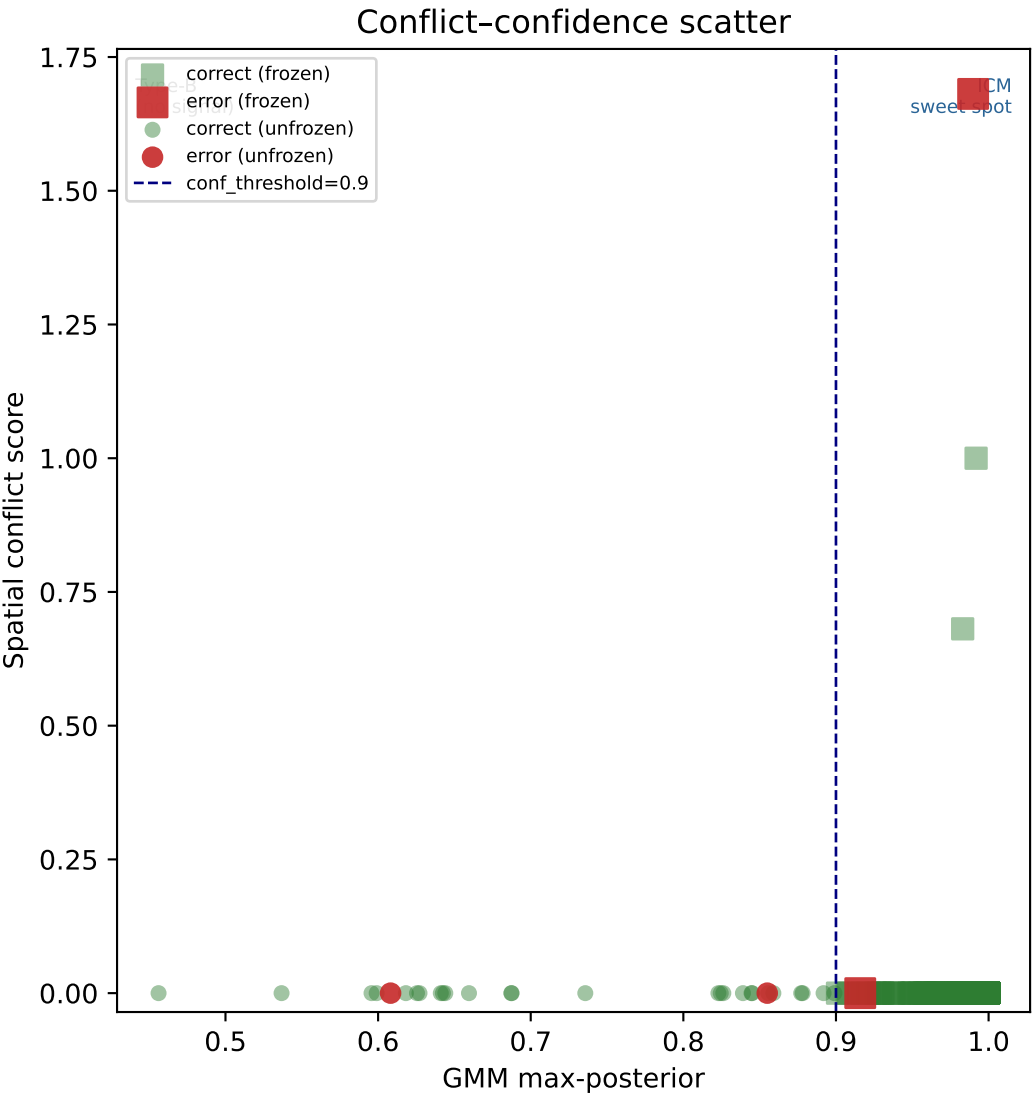
Run summary

MRF Run Summary	
ARI:	0.9799
errors:	4 / 490
violations:	16 → 2
n_changed:	10
n_iters:	2
n_frozen:	462
Parameters:	
lam:	20.0
tau_low:	0.3
tau_high:	0.4
radius:	7.059097824976937
conf_thr:	0.9

elongated — GMM + MRF (signed log-ramp a=10 + H1 th=0.85) ARI=0.980 violations 16 -> 2 changed=10 iters=2 force_unfrozen=0



elongated — MRF diagnostics (signed log-ramp + H1)



Run summary

MRF Run Summary	
ARI:	0.9799
errors:	4 / 490
violations:	16 → 2
n_changed:	10
n_iters:	2
n_frozen:	462
Parameters:	
lam:	20.0
tau_low:	0.3
tau_high:	0.4
radius:	7.059097824976937
conf_thr:	0.9